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Preprint in Management Science · March 2022

DOI: 10.1287/mnsc.2021.4257

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**Dynamic Pricing for New Products Using a
Utility-Based Generalization of the Bass Diffusion Model**

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October 14, 2021

Forthcoming at Management Science

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Dynamic Pricing for New Products Using a Utility-Based Generalization of the Bass Diffusion Model

Abstract

The Bass Model (BM) has an excellent track record in the realm of new product sales forecasting. However, its use for optimal dynamic pricing or advertising is relatively limited since the Generalized Bass Model (GBM), which extends the BM to handle marketing variables, uses only percentage changes in marketing variables rather than their actual values. This restricts the GBM's prescriptive use, for example, to derive the optimal price path for a new product, conditional on an assumed launch price, but not the launch price itself. In this paper, we employ a utility-based extension of the BM, which can yield normative prescriptions regarding both the introductory price, as well as the price path after launch, for the new product. We offer two versions of this utility-based diffusion model, namely the *Bass-Gumbel Diffusion Model (BGDM)* and the *Bass-Logit Diffusion Model (BLDM)*, the latter of which has been previously used by Krishnan, Seetharaman and Vakratsas (2012). We show that both *BGDM* and *BLDM* handily outperform the GBM in forecasting new product sales using empirical data from four product categories. We discuss how to estimate *BGDM* and *BLDM* in the absence of past sales data. We compare the optimal pricing policy of the *BLDM* with the *GBM* and derive optimal pricing policies that are implied by the *BLDM* under various ranges of model parameters. We illustrate a dynamic pricing approach that allows managers to derive optimal marketing policies in a computationally convenient manner; and extend this approach to a competitive, multi-product case.

Keywords: Dynamic Pricing, Optimal Pricing, Prescriptive Analytics, New Products, Diffusion Model, Bass Model (BM), Generalized Bass Model (GBM), New Product Sales Forecasting.

Introduction

The Bass Model (BM, henceforth), first proposed by Bass (1969), is the best-known new product diffusion model in the marketing literature. The BM has well served the managerial purpose of new product sales forecasting in thousands of industrial applications over the past 50 years (successful applications include forecasting market adoptions of DirecTV, XM-Sirius Satellite Radio, Microsoft Windows Operating Systems, etc.). The appealing aspect of the BM is its parsimony. Using just two parameters -- the innovation coefficient (p) and the imitation coefficient (q) -- the BM fits a wide range of observed adoption patterns, which reflect two influences on market adoption of the new product: the effects of external media and interpersonal communications (or “word of mouth”). While pre-launch sales forecasting has indeed been its main field of use, another important managerial purpose for the BM is to inform the development of optimal marketing policies for the new product, e.g., dynamic pricing, advertising, etc. (see Peres, Muller and Mahajan 2010 for an extensive review of the BM).

With regard to the above-mentioned managerial purpose of designing optimal marketing policies for new products, it is necessary for the BM to incorporate the effects of marketing variables on new product diffusion. Taking this view, the Generalized Bass Model (GBM, henceforth), which is an extension of the BM to handle marketing variables such as price, advertising, etc., was proposed by Bass, Krishnan and Jain (1994). The GBM nests the BM as a special case.¹ In fact, the GBM reduces to the BM when either the new product’s marketing mix is constant over time or when each marketing mix variable changes by a constant percentage amount from one period to the next. For example, suppose price is the only marketing variable of interest, if price always decreases by x % from one period to the next, then the BM and the GBM would do equally well in terms of fitting the observed sales pattern of the new product even though the BM ignores the effects of price. The reason for this is that the GBM overlays the effects of the marketing mix within the BM by parameterizing the diffusion speed in a given period to depend on only the percentage

¹ Several other models of new product adoption have been proposed in the literature to handle the effects of marketing variables. However, unlike the GBM, they typically do not reduce to the BM as a special case.

changes in the marketing mix variables from the previous period to the current period, and not the actual current values of the marketing mix variables. Therefore, if the percentage change in price is constant from one period to the next, the price change covariate is perfectly collinear with time, making it unnecessary to explicitly incorporate the effect of the price change in the model from the standpoint of forecasting sales.

The above-mentioned operationalization of marketing mix effects, using percentage changes under the GBM, makes it necessary to make additional assumptions before rendering the GBM usable for optimal price policy planning. For example, Krishnan, Bass and Jain (1999) make a specific parametric assumption about how the diffusion speed of the new product additionally depends on the absolute price of the new product. This parametric assumption is not, nor can be, validated using empirical data. The assumption is only made for the sake of mathematical necessity to solve the dynamic optimization problem characterizing the new product's optimal pricing policy. Krishnan and Jain (2006) make a similar untested assumption about how the diffusion speed depends on the absolute advertising level to derive the optimal advertising policy for a new product. Even after making such untested assumptions, the GBM can be used to derive only the optimal price path or advertising path for the new product after launch, conditional on an assumed launch price or advertising levels, but not the optimal launch price or advertising level themselves (which must be figured out using other means, such as a consumer survey, as discussed in Krishnan, Bass and Jain 1999). Furthermore, Fruchter and Van den Bulte (2011) explicitly show that different assumptions about how absolute levels of the marketing mix are incorporated in the GBM can lead to vastly different optimal marketing policies. On that basis, Fruchter and Van den Bulte (2011) argue that the GBM is ill-suited for normative marketing planning. In sum, therefore, the GBM, while being an excellent predictive model for sales forecasting, has a serious shortcoming as a prescriptive model for marketing policy planning. This shortcoming has been previously recognized in the literature (see, for example, Bass, Jain and Krishnan 2000).

In this paper, we employ an alternative extension of the BM, which is utility-based (unlike the GBM) and does not suffer from the prescriptive weakness of the GBM. Built on the foundations of consumer utility maximization, this utility-based diffusion model allows the adoption rate for the new

product to depend on the absolute levels of the marketing mix structurally.² Therefore, the diffusion model we use can yield, for example, normative prescriptions regarding both the introductory price and the optimal price path after launch for the new product. In addition to the marketing mix, the diffusion model can also incorporate market characteristics, category characteristics, and macro-economic variables, if available, to affect the consumer's utility function. We offer two versions of the diffusion model, namely the *Bass-Gumbel Diffusion Model (BGDM)* and the *Bass-Logit Diffusion Model (BLDM)*. The *BGDM* version is new to the literature and nests the BM as a special case and reduces to the BM when consumers are unresponsive to the marketing mix. On the other hand, the *BLDM* version – which has been previously estimated on SUV adoption data by Krishnan, Seetharaman and Vakratsas (2012) – does not nest the BM as a special case. However, the hazard of the *BLDM* mimics the parametric shape of the BM hazard very closely; therefore, in a practical sense, it *almost* nests the BM as a special case. Since product managers are often interested in figuring out the launch price and advertising strategy for their new products, and the GBM cannot be used to derive launch strategies for the marketing mix (as explained above); hence, both the *BGDM* and *BLDM* offer a practical benefit in that regard.

Using empirical data from 4 product categories (that have been previously used by Bass, Krishnan and Jain 1994 and Jiang, Bass and Bass 2006), we show that both *BGDM* and *BLDM* handily outperform the GBM and BM in forecasting new product sales both in sample and out of sample, with the *BLDM* outperforming the *BGDM*. On the basis of these findings, we propose that even if one's purpose is to only forecast new product sales, and not optimize the new product's marketing mix, both *BGDM* and *BLDM* are still to be preferred to the GBM. In addition, both *BGDM* and *BLDM* predict the peak time of adoptions more accurately than the GBM.

We illustrate the prescriptive consequences of the utility-based diffusion model for new product pricing, in terms of deriving both the optimal launch price and the optimal price path for the new product after product launch. Using a computationally convenient dynamic pricing approach, we compare the

² Utility-based models of new product adoption have been previously proposed in the literature. However, Bass, Krishnan and Jain (1994) show that the GBM consistently outperforms these models in predicting sales outcomes.

dynamic price path yielded by the *BLDM* versus that of the *GBM*. Finally, we extend this pricing approach to handle the competitive, multi-product context. Our prescriptive analytics solution will be useful to practitioners for new product planning decisions, especially given the ubiquitous global use of the BM for new product sales forecasting and characterizing internal versus external influences on new product adoption.

The rest of the paper is organized as follows. In section 2, we present a brief literature review. In section 3, we first derive the widely used BM as a utility-based diffusion model. Next, we illustrate how to incorporate the marketing mix variables into this utility-based diffusion model. Then, we lay out two different versions of the diffusion model, namely the *Bass-Gumbel Diffusion Model (BGDM)* and the *Bass-Logit Diffusion Model (BLDM)*, the latter of which has been previously estimated by Krishnan, Seetharaman and Vakratsas (2012). In section 4, we present the empirical results from estimating *BGDM*, *BLDM*, and two comparison models (BM, GBM), using new product sales data for four different product categories. In section 5, we discuss how to derive the optimal pricing policy for the new product, given the parameters of the utility-based diffusion model. We present a computationally convenient solution to the firm's dynamic optimization problem, which is easy for managers to use. In addition, we present the optimal pricing policies for a new product that are implied by the *BLDM* under various ranges of the model parameters. Then, we illustrate how to extend the dynamic pricing approach to a competitive, multi-product context. Section 6 concludes with directions for future research.

Literature Review

A number of researchers in the 1980s studied optimal dynamic pricing for new products using analytical models of dynamic pricing (Bass and Bultez 1982, Kalish 1983, Kalish and Lilien 1983, Thompson and Teng 1984, Dockner and Jorgensen 1988). These models relied on assumptions regarding consumer adoption patterns for new products. Subsequent research in the 1990s focused on explicit extensions of the BM, by then a workhorse empirical model for new product sales forecasting, to estimate

the effects of price on new product adoption (Jain and Rao 1990, Bayus 1992, Bass, Krishnan and Jain 1994). The extension that became most well accepted in the marketing literature is the GBM of Bass, Krishnan and Jain (1994), which has subsequently not only been used in several industry consulting applications but also built in to commercially distributed diffusion software such as DecisionPro's MEXL, Number Analytics etc. Given the wide acceptance of the GBM in both research and practice, Krishnan, Bass and Jain (1999) analytically derived the optimal dynamic pricing implications of the GBM. In doing this, Krishnan, Bass and Jain (1999) treated the introductory price of the new product as an exogenously pre-specified constraint to render the optimization problem solvable. In this research, we employ a utility-based extension of the BM, which does not warrant such a constraint for price optimization. We offer two versions of this utility-based diffusion model. Under both versions, the introductory price for the new product can be explicitly optimized rather than assumed.

Song and Chintagunta (2003) and Melnikov (2012) propose and estimate utility-based models of consumer adoption of durable goods. Unlike the utility-based model that we use, their models are not based on the BM. However, the implied aggregate adoption patterns yielded by their models could resemble the S-shaped adoption curve of the BM under some conditions. The aggregate adoption curve is derived as an endogenous consequence of the choices of individual consumers who are both heterogeneous in terms of their price sensitivity and forward-looking in terms of holding future price expectations for the product. In other words, the S-shaped temporal pattern in aggregate adoptions arises when product prices decrease over time and because consumers with different price sensitivities buy at different points of time during the product life cycle. In our setting, however, consumers are myopic (i.e., do not hold future price expectations) but respond to previous cumulative adopters on account of "word of mouth" effects, consistent with the behavioral foundations of the BM. In fact, in one of its versions, our model mathematically reduces to the BM as a special case, while the utility-based models of Song and Chintagunta (2003) and Melnikov (2012) would not. Further, those two papers do not study optimal dynamic pricing for a new product, which is the focus of our research.

Krishnan, Seetharaman and Vakratsas (2012) study consumer adoption of SUVs using a two-stage model of demand, where the first-stage model pertains to category adoption, and the second-stage pertains to brand choice. The first-stage model of category adoption is the *BLDM*, while the second-stage model of brand choice is a Multinomial Logit (MNL). Given our partial overlap with the past research of Krishnan, Seetharaman and Vakratsas (2012), what distinguishes our research from theirs? We note four contributions: First, while Krishnan, Seetharaman and Vakratsas (2012) estimate the *BLDM*, they do not estimate the GBM on SUV adoptions data. Therefore, the empirical usefulness of the *BLDM* relative to the GBM remains unknown. In this paper, we explicitly compare the empirical performance of the two models – *BLDM* versus GBM – on various predictive metrics across multiple datasets and show that the *BLDM* vastly outperforms the GBM. We also show that the *BLDM* better recovers the effects of price on new product adoption. Second, we offer two versions of the utility-based diffusion model, *BGDM* (that is new to the literature and nests the original BM as a special case) and *BLDM*, and compare their empirical performance. The *BLDM* outperforms the *BGDM* in terms of sales predictions and is, therefore, a more preferred version. Third, we illustrate how to extend the utility-based category adoption model to handle multiple brands in the category. On the other hand, Krishnan, Seetharaman and Vakratsas (2012) model brands’ market shares separately using a second-stage MNL model. Fourth, and most importantly, we study the normative usefulness of the utility-based diffusion model vis-à-vis the GBM for marketing mix optimization. Unlike the GBM, which suffers from the weakness of being unable to provide a prescription for the optimal launch price (as shown by Fruchter and Van den Bulte 2011), the utility-based diffusion model, when used as an input to a price optimization algorithm, yields an optimal launch price, as well as an optimal price path, for the new product. Such price optimization was not undertaken by Krishnan, Seetharaman and Vakratsas (2012). We show how to undertake the price optimization using an easy-to-implement numerical optimization algorithm (recently developed by Cosguner, Chan and Seetharaman 2018). This is the main contribution of this research.

In sum, our key research contribution lies in enabling marketing mix optimization – yielding an optimal price path – for a new product using a predictive model of new product adoption that more

accurately forecasts new product sales than existing models. We use a recently proposed numerical dynamic optimization technique for this purpose.

Proposed Utility-Based Generalization of the Bass Diffusion Model

We develop our model in four stages. First, we present the BM and its associated Non-linear Least Squares (NLS) estimation procedure for discrete-time data, first proposed by Srinivasan and Mason (1986), and now commonly used to estimate the BM by researchers and practitioners. Second, we show how to derive the discrete-time BM as a utility-based diffusion model. Third, we extend the utility-based BM to handle marketing variables to obtain the first version of the diffusion model, the *Bass-Gumbel Diffusion Model (BGDM)*. Fourth, we present the second version of the diffusion model, the *Bass-Logit Diffusion Model (BLDM)*. We discuss both the difference between *BGDM* and *BLDM* and the difference between *BGDM / BLDM* and the GBM.

Step 1: Bass Model (BM)

According to the BM (Bass 1969), in a market size of M consumers for a new product, the likelihood that a consumer will adopt the new product at time t , given that the consumer has not yet adopted, is given by the following equation.

$$\frac{f(t)}{1 - F(t)} = p + qF(t), \quad (1)$$

where p and q represent the coefficients of innovation and imitation, respectively, $F(t)$ is the cumulative distribution function (cdf) and $f(t)$ is the probability density function (pdf) characterizing the random variable t , which represents the consumer's time to adoption for the new product. Assuming $F(0) = 0$ and solving the differential equation (1) yields the following.

$$F(t) = \frac{1 - e^{-(p+q)t}}{1 + \frac{q}{p} e^{-(p+q)t}}. \quad (2)$$

Since new product sales data are commonly available in discrete time intervals (such as quarters or years), the discrete-time version of the BM is commonly estimated using the non-linear least squares procedure, first proposed by Srinivasan and Mason (1986), by minimizing the following criterion function.

$$SSE = \sum_{t=1}^T [\hat{N}(t) - N(t)]^2, \quad (3)$$

where T stands for the total number of periods for which new product sales data are available, $N(t)$ represents the observed sales of the new product in period t , and $\hat{N}(t)$ represents the predicted sales of the new product in period t , and is given by

$$\hat{N}(t) = M * [F(t) - F(t-1)]. \quad (4)$$

Minimizing the criterion function in equation (3) yields estimates of p , q , and M for the new product. Sales forecasts can then be generated for the new product for any period t , whether in sample or out of sample, using equation (4) above. When no past sales data are available on the new product, such as prior to the launch of the new product, sales data of comparable products, called “analogs,” are first used to estimate p and q for the analogs, after which a weighted average (where the weights represent similarities of the analogs to the new product) of the analogs’ p and q values is taken to represent p and q for the new product. M is estimated separately by combining information on product purchase intent obtained from a consumer survey with census data using a chain-ratio method. These estimates of p , q , and M for the new product are then used to forecast new product sales, again using equation (4) above. Typically, practical applications of the BM rely on analog-based sales forecasting prior to product launch, then use Bayesian updating of p , q and M after sales data begin to appear for the new product and eventually rely only on historical sales data of the new product to forecast future sales.

Step 2: Deriving the Discrete-Time BM as a Utility-Based Diffusion Model

In a market size of M consumers for a new product, suppose the utility of a consumer for the new product at time t is given by the following equation.

$$U_t = \ln \left[\ln \left\{ \frac{1 - F(t-1)}{1 - F(t)} \right\} \right] + \varepsilon_t, \quad (5)$$

where the first term on the right-hand side, $\ln \left[\ln \left\{ \frac{1 - F(t-1)}{1 - F(t)} \right\} \right]$, represented by V_t for short, represents the *deterministic* attractiveness³ of the new product to the consumer at time t , $F(t)$ represents the cdf given in equation (2), and ε_t represents an unobserved (by the researcher) random component that captures other influences on the consumer's utility for the new product. If one assumes that the random component ε_t follows a Gumbel distribution with location parameter 0 and scale parameter 1, it can be shown that the consumer's probability of buying the new product at time t , given that the consumer has not yet bought the new product, also called the *discrete hazard function* characterizing the consumer's time to adoption for the new product, is given by the following equation (Seetharaman 2004).

$$\text{Pr}_t = 1 - e^{-e^{V_t}}, \quad (6)$$

which, in turn, can be rewritten as follows.

$$\text{Pr}_t = \frac{F(t) - F(t-1)}{1 - F(t-1)}, \quad (7)$$

which implies that the consumer's probability of not buying the new product at time t , given that the consumer has not yet bought the new product, also called the *discrete survivor function* characterizing the consumer's time to adoption for the new product, is given by the following equation.

$$1 - \text{Pr}_t = \frac{1 - F(t)}{1 - F(t-1)}. \quad (8)$$

The consumer's unconditional likelihood of buying the new product at time t , also called the *probability mass function* characterizing the consumer's time to adoption for the new product, depends on the consumer's discrete hazard and discrete survivor functions, as follows.

$$L_t = [1 - \text{Pr}(1)] * [1 - \text{Pr}(2)] * \dots * [1 - \text{Pr}(t-1)] * \text{Pr}(t), \quad (9)$$

³ The *deterministic attractiveness* term, which reflects the consumer's intrinsic utility for the product, represents the *discrete-time baseline hazard* of adoption (Krishnan, Seetharaman and Vakratsas 2012). It is strictly increasing over time, which means that a consumer's conditional probability of adopting the product, given that adoption has not happened yet, increases over time.

which can be rewritten, using equations (7) and (8), as follows.

$$L_t = \frac{1-F(1)}{1-F(0)} * \frac{1-F(2)}{1-F(1)} * \dots * \frac{1-F(t-1)}{1-F(t-2)} * \frac{F(t)-F(t-1)}{1-F(t-1)}, \quad (10)$$

which reduces to the following.

$$L_t = F(t) - F(t-1). \quad (11)$$

This implies that the predicted sales of the new product in period t is given by

$$\hat{N}(t) = M * [F(t) - F(t-1)]. \quad (12)$$

Equations (4) and (12) are identical. This establishes that the discrete-time version of the BM can be derived as a utility-based diffusion model, as formulated in equation (5). This equivalence between a utility-based discrete choice model and the BM has not been explicitly recognized in the marketing literature (except for Krishnan, Seetharaman and Vakratsas 2012 where it is tacit in their model derivation). Intuitively, equation (5) can be understood as follows: Since $F(t)$ is a monotonic increasing function of t , i.e., $F(t) > F(t-1)$, it follows that $\ln \left[\ln \left\{ \frac{1-F(t-1)}{1-F(t)} \right\} \right]$ is a monotonic increasing function of t . This means that the deterministic attractiveness of the new product to the consumer, V_t , is an increasing function of t , which, in turn, implies that the discrete hazard, Pr_t , is also an increasing function of t as per the continuous-time BM represented in equation (1).

Step 3: Bass-Gumbel Diffusion Model (BGDM)

In a market size of M consumers for a new product, suppose the utility of a consumer for the new product at time t is given by the following equation.

$$U_t = \ln \left[\ln \left\{ \frac{1-F(t-1)}{1-F(t)} \right\} \right] + Price_t * \beta + \varepsilon_t, \quad (13)$$

where $Price_t$ represents the price of the new product at time t and β represents the price sensitivity of the consumer, and the remaining terms are as in equation (5). Under the same Gumbel distributional assumption for ε_t as in equation (5), this implies that the consumer's probability of buying the new product at time t ,

given that the consumer has not yet bought the new product, also called the *discrete hazard function* characterizing the consumer's time to adoption for the new product, is given by the following equation.

$$\Pr_t = 1 - e^{-e^{W_t}}, \quad (14)$$

where

$$W_t = \ln \left[\ln \left\{ \frac{1 - F(t-1)}{1 - F(t)} \right\} \right] + \text{Price}_t * \beta. \quad (15)$$

The consumer's unconditional likelihood of buying the new product at time t is as follows.

$$L_t = \left[\prod_{u=1}^{t-1} 1 - \Pr(u) \right] * \Pr(t), \quad (16)$$

and the predicted sales of the new product in period t is, therefore,

$$\hat{N}(t) = M * L_t. \quad (17)$$

The criterion function in equation (3) is minimized to estimate the parameters -- p , q , M , β -- of this diffusion model, that we call the *BGDM*. This model reduces to the traditional BM when $\beta = 0$. In this sense, the *BGDM* generalizes the BM (as does the GBM) by nesting the BM as a special case.

What if no past sales data are available on the new product, such as prior to the launch of the new product? ⁴ In that case, sales data of comparable products, called “analogs,” can be first used to estimate p and q for the analogs, after which a weighted average, where the weights represent similarities of the analogs to the new product, of the analogs' p and q values is taken to represent p and q for the new product. This is the standard approach taken to estimating p and q in the absence of past sales data under the BM (see Ofek 2008 for details). The remaining parameters, M and β , can be estimated using a *discrete choice*, also called *choice-based conjoint*, survey (see Louviere, Hensher and Swait 2000), which is conducted as follows. First, the new product prototype, in real or virtual form, is shown to respondents so that they fully understand the functional features of the new product. Second, respondents undertake 18 choice tasks,

⁴ We thank an anonymous reviewer for raising this important question about the practical usability of the diffusion model in the absence of historical sales data.

divided in to 6 sets of 3 choice tasks each. Each set of 3 choice tasks represents a subsequent year, when compared to the previous set of 3 choice tasks, since hypothetical introduction of the new product. For example, in the first 3 choice tasks, the respondent is told that the new product has just been launched during the current year; in the next 3 choice tasks, the respondent is told that the new product was launched during the previous year; in the last set of 3 choice tasks, the respondent is told that the new product was launched 6 years prior to the current year. The respondent is also asked to assume that he or she has not yet adopted the new product while responding to each choice task. Within each choice task, which represents a specific period t since launch, the respondent is shown a particular version of the new product, defined in terms of how long ago it was introduced (as explained above), as well as two other specific attributes: (1) installed base of existing adopters ($M * F(t)$), and (2) price of the new product ($Price_t$). The respondent is then asked for their likelihood (Pr_t) of adopting that version of the new product.

In sum, in the discrete choice survey, as discussed above, the 18 versions of the new product differ in terms of $\ln \left[\ln \left\{ \frac{1-F(t-1)}{1-F(t)} \right\} \right]$, and $Price_t$, which are the two covariates on the right-hand side of equation (13). The respondent's likelihood of adoption for each version of the new product serves as the empirical measure of Pr_t , which is on the left-hand side of equation (14). We now have a survey-based counterpart dataset to proxy for the absent historical sales data, which can be used to estimate the parameters β and M of the *BGDM*, using equations (15) - (17). In this manner, one can estimate the parameters -- p , q , M , β -- of the *BGDM* in the absence of historical sales data on the new product.

Step 4: Bass-Logit Diffusion Model (BLDM)

Suppose the utility function in equation (13) still applies, but we assume that the random component, ε_t , follows the Logistic (rather than Gumbel) distribution with location parameter 0 and scale parameter 1. This implies that the consumer's probability of buying the new product at time t , given that the consumer has not yet bought the new product, also called the *discrete hazard function* characterizing the consumer's time to adoption for the new product, is given by the following equation.

$$\Pr_t = \frac{e^{W_t}}{1 + e^{W_t}}, \quad (18)$$

where W_t is given by equation (15). We call this model the *BLDM*. This model has been previously estimated on SUV adoption data by Krishnan, Seetharaman and Vakratsas (2012). The only feature that distinguishes the *BLDM* from the *BGDM* is the assumption about the random component, ε_t . Equations (16) and (17) still apply and the criterion function in equation (3) is still minimized to estimate the parameters of the *BLDM*. Unlike the *BGDM*, the *BLDM* will not reduce to the traditional BM when $\beta = 0$. However, it will reduce to a logistic counterpart of the discrete-time BM instead. Since the logit model is a workhorse random utility model in the marketing literature, inspired by the seminal work of Luce (1958) and a highly cited empirical application by Guadagni and Little (1983), we offer this version as an alternative formulation of the utility-based diffusion model even though it does not nest the BM as a special case.

As a comparison model, which presents an alternative way of incorporating marketing variables within the BM (as opposed to the derived utility-based diffusion model), we present the GBM below. The cdf characterizing the consumer's time to adoption of the new product under the GBM specification is,

$$F(t) = \frac{1 - e^{-(p+q)X(t)}}{1 + \frac{q}{p} e^{-(p+q)X(t)}}, \quad (19)$$

where

$$X(t) = t + \beta \ln \left(\frac{\text{Price}_t}{\text{Price}_0} \right), \quad (20)$$

where Price_0 is the launch price for the new product (which needs to be assumed to be equal to Price_1 in the estimation of the GBM; even more importantly, the value of Price_0 needs to be assumed, rather than explicitly optimized, while deriving the optimal pricing policy using the GBM, as explained earlier). We estimate the GBM using equations (3) and (4) as well.

Model Discussion

The utility-based diffusion model that we have derived, on account of having a utility foundation, makes it natural to incorporate marketing variables in the usual additive manner within the consumer's utility function (see equation (13)). The GBM, on the other hand, does not operationalize marketing variables within the BM framework using a behavior-theoretic foundation. It simply embeds the effects of marketing variables on the adoption hazard using a multiplicative function. In fact, it is not possible to easily derive the discrete-time GBM as a utility-based diffusion model (in the same manner as we derive the discrete-time BM as a utility-based diffusion model). In addition to the marketing mix, the utility-based diffusion model can also incorporate market characteristics, category characteristics and macro-economic variables, if available, to affect the consumer's utility function by including them, for example, in linear additive form in equation (13). Some of these variables can be categorical, and some could represent fixed effects if necessary. It is unclear how one would include such variables, especially if they are time-invariant, in a behavior-theoretic manner within the GBM. One could also use control function approaches to correct for possible endogeneity of marketing variables (if suitable instruments are available) by including the fitted residuals from the instrumental variable regressions as additional explanatory variables in the consumer's utility function. Endogeneity correction in the GBM, however, is a non-trivial task.

It is useful to note that Bass, Jain and Krishnan (2000) propose a discrete-time Proportional Hazard Model (PHM) as an alternative diffusion model to the GBM. However, their proposed PHM does not reduce to the BM as a special case. Since the BM has long-standing precedent in the diffusion literature and has enjoyed great success in new product forecasting applications, we retain the BM within the *BGDM* as a special case. In the absence of marketing variables, the *BGDM* reduces to the traditional BM. Our goal is to exploit the excellent forecasting properties of the adoption hazard that is implied by the BM, while also additionally modeling the effects of marketing variables so that our model can be used for prescriptive decision-making. An added benefit of the utility-based diffusion model is that it can handle multiple, competing new products as we illustrate in our "Optimal Dynamic Pricing Policies" section, while the GBM cannot. This makes the utility-based diffusion model more convenient for practical use since many new

product categories involve multiple, competing new products diffusing simultaneously in the market (e.g., iPhone, Samsung Galaxy, LG Tracfone etc.). Last, but not least, since the utility-based extension of the BM makes specific parametric assumptions (Gumbel or Logistic) about the utility function, the model can also be estimated using maximum likelihood (rather than least squares), which makes it easy to incorporate random coefficients, using discrete or continuous mixture distributions, to reflect unobserved heterogeneity across consumers.

It is useful to note that the Gumbel (for BGDM) and Logistic (for BLDM) distributions of error terms yield convenient closed-form expressions for consumer adoption probabilities (see equations (14 and (18)). Our choice of the Logistic error distribution is consistent with an extensive literature in marketing, statistics and econometrics on discrete choice models. Our choice of Gumbel distribution allows us to establish a random utility foundation for the widely used BM (as shown earlier). Furthermore, there is rich precedence in the econometrics literature for the choice of Gumbel errors in duration time models (see, for example, Heckman and Singer 1984, Meyer 1990, Han and Hausman 1990 for applications to unemployment durations). Extreme value distributions are the limiting distributions for the maximum of a very large collection of random observations from the same arbitrary distribution. Gumbel (1958) showed that for any well-behaved initial distribution of unobserved random errors, the Gumbel distribution is sufficient as such a limiting distribution. In the case of duration time models, extreme value distributions for the maximum apply. For example, if there are n unobserved influences on consumer adoption time, and consumer utility for the product crosses the adoption threshold as long as the maximum of these influences becomes sufficiently large, then extreme value theory implies that independent of the choice of component errors, the maximum will approach Gumbel as n becomes large. This provides a behavioral basis for the Gumbel error term in the BGDM. As far as the BLDM is concerned, because the Logistic error term can be shown to be a difference of two Gumbel error terms, if one views not only the random utility of the BLDM as Gumbel, but also the adoption threshold of the BLDM as Gumbel (instead of deterministic), then the BLDM can also be motivated as arising from a Gumbel specification of error terms.

Benefit of the *BGDM* and *BLDM* over the *GBM* in the Case of No Past Data

In the absence of historical sales data, such as prior to the launch of the new product, one can estimate the parameters of the *BGDM* or *BLDM*, including specifically the price sensitivity parameter (β), using discrete choice survey data, as discussed earlier. Such estimation is facilitated by the fact that choice-based conjoint relies on random utility primitives, as do *BGDM* and *BLDM*. However, under the *GBM*, which does not have a random utility foundation, it is unclear how one would take the market price sensitivity that is estimated using a conjoint survey and convert it to yield the price parameter of the *GBM*. In other words, the *GBM* appears unusable in the absence of historical sales data. In this sense, the strength of *BGDM* and *BLDM* over the *GBM* becomes even greater in the absence of past sales data.

Empirical Results

We have discussed the theoretical merits of the utility-based diffusion model we derive. Now we study how this model fares, relative to comparison models (BM and GBM), in explaining actual new product sales data. For this purpose, we employ annual new product adoptions data, along with annual average prices, pertaining to four categories of durable goods: Color TV, Air Conditioner, Clothes Dryer and Freezer. The reason for choosing these categories is that three of these categories (except freezer) were used in the first empirical application of the GBM by Bass, Krishnan and Jain (1994). All four categories were used in a recent application of the GBM by Jiang, Bass and Bass (2006). We use the exact same datasets that have been used in these previous papers.

We compare the empirical performance of the two versions of the utility-based diffusion model – *BGDM* and *BLDM* -- versus two comparison models, BM and GBM. The results of fit comparisons are given in Table 1. In terms of the fit criterion (SSE), we see that the *BLDM* does the best, while the *BGDM* does second best, in three out of four categories (AC, Dryer, Freezer). Not surprisingly, since it ignores the effects of prices, the BM does the worst in all 4 categories. Interestingly, in the only category (Color TV) where the GBM outperforms the utility-based diffusion model, its superior performance is only marginal (SSE of 402,514 for GBM versus 457,051 for *BGDM*), in sharp contrast to the utility-based model vastly

outperforming the GBM in the remaining three categories (for example, SSE of 89,550 for *BLDM* versus 294,659 for GBM in the Air Conditioners category). This reveals that the utility-based diffusion model is able to much better explain new product adoption patterns than the GBM.

[Insert Table 1 here]

It is possible that an in-sample fit criterion may exaggerate the superior empirical performance of the utility-based diffusion model versus the GBM and BM. For this reason, we additionally compare model performance using out-of-sample holdout predictions. For this purpose, we re-estimate all four diffusion models using only the first 80 % of the observations in the data (training sample), and then use the estimates to make predictions of sales in each of the last 20 % of observations in the data (holdout validation sample). Table 2 provides the results from the holdout validation comparison, period by period for each product category. The table shows that the predicted sales yielded by the *BLDM* comes closer to the actual sales in 11 out of the 12 cases, when compared to the predictions yielded by the other models. In the only case where *BLDM* does not yield the best prediction, the *BGDM* does the best. In other words, both versions of the utility-based diffusion model yield sales predictions that are closer to the actual sales data in the holdout validation sample in all cases when compared to the GBM or BM. This strongly vindicates the forecasting performance of the utility-based diffusion model. On the basis of these findings, we propose that even if one's purpose is to only forecast new product sales, and not optimize the new product's marketing mix, *BGDM* and *BLDM* must be preferred to the GBM.⁵

[Insert Table 2 here]

Next, we compare the prediction performance of the three diffusion models that incorporate price – *BLDM*, *BGDM*, GBM -- using k-fold cross validation. We do this in two ways: (1) Leave one out, and (2) Leave two out. In (1) we leave one observation out within each validation fold, and in (2) we leave two observations out within each validation fold. Given N observations in a category dataset, this results in N

⁵ We also perform holdout predictions using the first 60%, 70%, and 90% of the data as training samples, and find that the *BLDM* / *BGDM* outperform the BM / GBM in 75%, 92%, and 75% of the cases, respectively. Please see our Web Appendix A for the details of these additional holdout validations.

folds in (1) and $N*(N-1) / 2$ folds in (2). Within each holdout fold, we predict the outcome(s). Table 3 provides the results from the k-fold cross validation, reporting the percentage of times across all folds that each model is the “best” in terms of predicting validation outcomes. We find that one of the utility-based diffusion models – *BLDM* or *BGDM* – outperforms the GBM in a majority of the cases for each category, in both (1) and (2). For example, in the clothes dryer category, in the leave two out case, either *BLDM* or *BGDM* best predicts validation outcomes 72 % of the time while the GBM wins 28 % of the time. These results further highlight the practical usefulness of the utility-based diffusion models for prediction purposes.

[Insert Table 3 here]

We report the estimated parameters – innovation coefficient (p), imitation coefficient (q), market potential (M), and price sensitivity (β) – for each model in Table 4. We see that the estimated price sensitivity (β) is negative and significant for each category, as expected, under the *BGDM* and *BLDM*. This means that as price of the new product decreases (increases), its adoption speed accelerates (decelerates). However, under the GBM, the estimated price coefficient is insignificant in three out of four categories (except Color TV). This means that using the GBM, one could spuriously conclude that price does not significantly influence new product adoption even though our utility-based diffusion model may suggest otherwise. The estimated market potential (M) is lower under the GBM and BM than under the utility-based diffusion model in three out of four categories (except freezer). To investigate this issue further, we look at the re-estimated values of M in the training sample (i.e., first 80 % of observations), as opposed to the full sample. These are reported in Table 4(b). For all four categories, we note a systematic and large downward bias in the estimated values of M under the GBM and BM. This is consistent with the findings in Van den Bulte and Lilien (1997) that the estimated value of M goes systematically downward as one right-truncates the observed diffusion data. The two versions of the utility-based diffusion model, on the other hand, yield roughly similar estimates of M for two categories, and understated, albeit much less so than the GBM and BM, estimates for the other two categories. This highlights an additional benefit to using either the *BGDM* or *BLDM*, i.e., robustness to minor right truncations in the data.

[Insert Table 4 here]

It is difficult to directly interpret the estimated p and q coefficients in Table 4. Therefore, we convert these estimates to probability densities of adoption times (under the assumption that price stays at the introductory price, so that we are able to isolate the intrinsic adoption pattern, absent price effects), using the formulas in equations (11) and (16), for the BM / GBM and the *BGDM* / *BLDM*, respectively. First, we note that the estimated probability density is almost identical between the BM and GBM, and between the *BGDM* and *BLDM*. However, there are significant differences between the probability densities yielded by the *BGDM* / *BLDM* versus those yielded by the BM / GBM. For two categories (Color TV and Air Conditioner), the *BGDM* / *BLDM* implies a much longer, and therefore slower, adoption cycle, versus the BM / GBM (“fatter tails”). For one category (Freezer), we find the reverse, with the BM / GBM implying a longer adoption cycle. For one category (Clothes Dryer), the estimated probability densities do not differ by much across the four models. While there is no systematic pattern in these differences, it is interesting to see that the utility-based diffusion model yields a different intrinsic adoption pattern (which is still formulated as following a Bass curve when marketing variables are absent) when compared to the BM or GBM. Therefore, beyond the superiority in forecasting performance discussed earlier, *BGDM* / *BLDM* yield different intrinsic adoption patterns than the GBM. In other words, the descriptive content of *BGDM* / *BLDM* is different from that of the GBM.

Some comments are in order here. It is useful to note that the *BGDM* without price is mathematically equivalent to the BM and, therefore, would yield the same estimates of p , q , M as the BM. Therefore, the only reason that the estimates of p , q , M yielded by the *BGDM* are different from those yielded by the BM is the inclusion of price in *BGDM*. In other words, the observed differences in the estimates of p , q , M between the *BGDM* and BM is on account of *omitted variable bias* in the BM. That said, it is interesting to note that the estimates of p , q , M yielded by GBM do not differ that much from those yielded by the BM. In other words, the effects of omitted variable bias do not seem to show up when comparing the GBM to the BM. Why is this the case? Our explanation is that since the GBM incorporates the effect of price by linearly adding the percentage change in price (scaled by the price coefficient) to

calendar time, there is little to no consequence of ignoring price in the GBM when percentage change in price is almost collinear with the time variable. This, in turn, makes the estimated parameters p , q and M immune to whether or not price is included in the GBM. In *BGDM* / *BLDM*, however, price is included as an absolute measure, rather than as a percentage change, within the utility function. Additionally, time is represented non-linearly in the utility function using the $\ln [\ln (.)]$ term, which means that there is no collinearity between how price versus elapsed time are allowed to influence adoption. Therefore, the *BGDM* / *BLDM* successfully identify the effects of price on adoption in all four categories, unlike the GBM which fails to identify significant price effects in three out of four categories.

[Insert Figure 1 here]

In order to further elucidate the substantive implications of the estimated p and q coefficients, we compute the predicted peak adoption time based on the estimated model parameters, and present them in Table 5. First, we note that the predicted peak adoption times from the *BGDM* / *BLDM* exactly agree with the observed peak adoption times in 3 out of 4 categories (Color TV, Air Conditioner, Clothes Dryer). In the fourth category (Freezer), the predicted peak adoption time is slightly (by one period) larger than the observed value. In sharp contrast, the predicted peak adoption time from the BM / GBM agrees with the observed peak adoption time in only one category (Color TV). In the second and third categories (Air Conditioner, Clothes Dryer), the BM / GBM overestimate the peak adoption times by two periods, while in the fourth category (Freezer), the BM / GBM underestimate the peak adoption time (by 2 periods for the BM, and by 4 periods for the GBM). In sum, the *BGDM* / *BLDM* are able to predict peak adoption times much more accurately than the BM / GBM. This further vindicates the managerial usefulness of the utility-based diffusion model. We also predict the “lifetime” of the adoption lifecycle for each product using the number of periods until the number of adoptions in a period falls to less than 10 (see Table 5). For 3 out of 4 categories, we see that the *BGDM* / *BLDM* predict a much longer adoption lifetime (1.5 – 2 times) than what is predicted by the BM / GBM. To the extent that peak adoption times are more accurately predicted by *BGDM* / *BLDM*, one must place more faith on their predictions of adoption lifetimes as well.

In Table 5, we also report the predicted number of adoptions during the predicted peak time of adoptions. For 2 out of 4 categories (Color TV, Air Conditioners), the predictions of *BGDM* / *BLDM* come closer to the observed numbers of adoptions, while for the remaining 2 categories (Clothes Dryer, Freezer), the GBM makes closer predictions. On average, across the 4 categories, *BGDM* / *BLDM* yield a lower prediction error than the BM / GBM. ⁶

[Insert Table 5 here]

Optimal Dynamic Pricing Policies

We have already shown in section 4 that *BGDM* / *BLDM* have excellent forecasting properties using empirical data. Now we derive the prescriptive pricing policies that are implied by the utility-based diffusion model under various ranges of model parameters. For this purpose, we assume that the innovator firm maximizes the revenue yielded by their product. In reality, it is more likely that the innovator firm maximizes profit, rather than revenue. However, we observe neither the firm’s actual marginal cost of producing the product, nor the dependence of such marginal cost on the quantity produced (“economies of scale”) and the firm’s production experience in the category (i.e., “experience curve effects”). Therefore, in the absence of a known marginal cost function characterizing the firm, we cannot correctly specify the firm’s profit function far less maximize it. For this purpose, we use revenue maximization to illustrate our price optimization approach. It is easy to use our pricing approach to handle profit maximization: one would simply replace the objective function to reflect profit rather than revenue. For these numerical simulations, we use a modified version of the dynamic optimization algorithm developed in Cosguner, Chan and Seetharaman (2018) and use the *BLDM* version -- which has been previously used by Krishnan, Seetharaman and Vakratsas (2012) -- of the utility-based diffusion model. In the following subsections, we

⁶ To illustrate the superior predictive performance of *BGDM* / *BLDM*, we compare them with earlier diffusion and proportional hazard models in literature. Please see our Web Appendix B for the details of this comparison exercise. In an additional analysis, we use two recent datasets and illustrate *BGDM* / *BLDM*’s superior performance as opposed to GBM. Please see our Web Appendix C for the details of this exercise.

discuss our dynamic optimization algorithm first, supply-side implications for an innovator firm next, and the extension of our dynamic pricing approach to handle multiple products last.

Numerical Optimization Algorithm to Solve an Innovator Firm's Dynamic Pricing Problem

The pricing objective of the innovator firm selling the new product is as follows.

$$\{\text{Price}_1, \text{Price}_2, \dots, \text{Price}_T\} = \operatorname{argmax}_{\sum_{t=1}^T \rho^{t-1} * \text{Price}_t * N_t}, \quad (21)$$

where ρ is the time discount factor, Price_t is the price of the new product at time t , and N_t is the expected sales at time t . In this setting, the revenue maximizer firm's decision for the current price affects not only the current period revenue, but also the revenue stream in the future through changing the remaining market potential for the new product. The relevant state variables (Z) for the innovator firm's dynamic pricing problem are, therefore, time and the remaining market potential, i.e., $Z_t = \{t, R_t\}$, where $R_t = M - \sum_{k=1}^{t-1} N_k$. In order to find the optimal pricing policy function for the new product, we adapt the dynamic optimization algorithm developed in Cosguner, Chan and Seetharaman (2018). First, we parameterize the firm's temporal pricing policy as a flexible function of the state variables, Z , as shown below.

$$\text{Price} = \widehat{\text{Price}}(Z|\theta), \quad (22)$$

where $\widehat{\text{Price}}(\cdot)$ denotes a flexible parametric approximation of the firm's pricing policy, and θ is a vector of parameters characterizing this parametric approximation. In order to find the true θ for a given set of diffusion parameters and the discount factor, i.e., $(p, q, M, \beta \text{ and } \rho)$, we numerically search for θ that maximizes the lifetime product revenue. This requires us to calculate the lifetime revenue for each candidate value of θ . To do that, we use the following forward simulation algorithm.

For each candidate value of θ :

- i. Start with the first period, i.e., $Z_1 = \{t = 1, R_1 = M\}$. Given Z_1 , calculate $\text{Price}_1 = \widehat{\text{Price}}(Z_1|\theta)$. Given Price_1 and Z_1 , calculate the first period sales, $N_1(\text{Price}_1, Z_1)$, using equation (4) or (17) for the BM / GBM or *BGDM* / *BLDM*, respectively.

- ii. Given $Z_2 = \{t = 2, R_2 = M - N_1\}$, calculate the second period price, $\text{Price}_2 = \widehat{\text{Price}}(Z_2|\theta)$. Given Price_2 and Z_2 , calculate the second period sales, $N_2(\text{Price}_2, Z_2)$, using equation (4) or (17).
- iii. Repeat (ii) until an iteration T such that $R_T = M - \sum_{k=1}^{T-1} N_k \approx 0$.
- iv. Calculate the discounted sum of revenues, i.e., $\sum_{t=1}^T \rho^{t-1} * \text{Price}_t * N_t$. That will be the lifetime revenue for the given policy function $\widehat{\text{Price}}(Z|\theta)$.

For our implementation, we use the second order polynomial approximation with interaction terms for the parameterized policy function (i.e., θ contains an intercept parameter, and slope parameters for t , R_t , t^2 , R_t^2 , $t*R_t$). Since the state transitions are deterministic, and the market is fully covered within a limited number of iterations, the simulation exercise becomes extremely fast. In other words, the user can recover the optimal path of prices within a few seconds using a programming environment such as R (please consult Web Appendix D for the sample R code). This is a computationally convenient solution to the firm's dynamic optimization problem which is easy for managers to use. We discuss the supply-side pricing implications next.

Optimal Dynamic Pricing Implications for the Innovator Firm

First, we report the optimal price paths, alongside the actual price paths (as observed in the data), for all four categories in Figure 2. The key take-aways of Figure 2 are as follows: First, the optimal price path is non-monotonic, first increasing and then decreasing over time, in each of the four categories. The non-monotonicity is most pronounced in the Color TV category, with an introductory price of \$580 quickly rising to \$850 by period 9, and then gradually decreasing to \$745 by period 25. The dynamic price path is most shallow in the Clothes Dryer category in the sense that price shows only a minor initial rise and subsequent fall over time.⁷ Plots like these, which represent dynamic optimal pricing policies, can be useful

⁷ Since the recovered price path is a function of estimated model parameters, which are associated with standard errors, we undertake sensitivity analysis to study the impact of model uncertainty. Web Appendix E reports the optimal price paths based on

to managers in innovator firms in terms of enabling related marketing decisions, such as advertising and sales planning, to be congruent with the prices. Second, consistent with the optimal price paths, we observe that actual price paths are non-monotonic – first increasing and then decreasing – in the Clothes Dryer and Freezer categories. However, they are steadily decreasing in the Color TV and Air Conditioner categories. In three out of four categories – Color TV, Clothes Dryer, Freezer – the actual prices are consistently lower than the actual prices. In the Air Conditioner category, however, the actual prices are higher than optimal prices in early periods, and lower in later periods.

One should not interpret the observed differences between optimal and actual price paths, especially with regard to their shapes – nonmonotonic optimal prices, versus monotonically decreasing actual prices – in two categories, as speaking to the face validity (or lack thereof) of our price optimization approach. Our intent in this research is not to descriptively explain the observed price patterns in the data. Instead, our goal is to provide an easy-to-implement normative algorithm for price optimization that uses the descriptively appropriate new product adoption model as an input. As discussed earlier, the actual prices in the data are most likely a consequence of profit maximization on the part of the innovator firm, while we wrongly assume revenue maximization in our numerical simulation. It would require conversations with industry managers who were managing product pricing in those categories during the early years of adoption in order for us to understand the (possibly time-varying) marginal cost function characterizing the product. Using that information and rerunning our numerical simulation using a suitably specified profit function would then allow us to derive the optimal price path for a profit maximization objective. That said, we provide a speculative explanation for the actual price path seen in our data: Suppose the seller experiences declining costs over time (“experience curve” effect), and we assume that their profit margin (m) increases over time as follows: $m = 1 - [0.9 / (t^{0.1})]$. Under this assumption, and additionally assuming

the estimated 95 % confidence intervals for p and q . The results suggest the following: (1) the 95% prediction bands are quite “tight” for p , which means that the impact of uncertainty in the estimated value of p on optimal prices is minimal, (2) the 95% prediction band is wider for q , but not worrisomely wide due to parameter uncertainty. For example, the maximum price over the dynamic price path varies from \$770 (lower bound) versus \$880 (upper bound) in the Color TV category, or a difference of \$110, on account of model uncertainty. In practice, the risk aversion and objective functions of managers will determine how they make use of this uncertainty in pricing prescriptions and its effect on the category profit.

that the seller maximizes profit (rather than revenue), we find that optimal prices decrease over time for various values of p and q .⁸ Therefore, one explanation for the decreasing temporal path of prices that are observed in the data is the existence of experience curve effects. Since we do not observe seller costs in our data, we are unable to empirically verify whether this explanation is indeed correct. In practice, however, sellers can use the suggested dynamic pricing approach to maximize profit (rather than revenue, as we do in this study). It is important to note that the suggested price optimization approach can yield different shapes – decreasing, increasing, flat, U-shaped, inverted U-shaped etc. -- for the optimal price paths depending on both the parameters of the new product adoption model, as well as the objective function being maximized by the firm.

[Insert Figure 2 here]

Next, we derive the optimal dynamic pricing policy that is implied by the GBM in order to understand differences in its pricing implications from the *BLDM*. For this purpose, we assume the introductory price ($Price_0$) to be the observed introductory price in the data (since the GBM requires introductory price to be exogenously pre-specified as discussed in the introduction section). Since the estimated effect of price under the GBM is significant only in the Color TV category, we perform this exercise in that category. The optimal price path implied by the GBM is shown in Figure 3 (alongside that implied by the *BLDM*). We note substantively significant differences in the pricing implications of the GBM when compared to those of the *BLDM*. Specifically, the *BLDM* suggests an increase in price of the product from \$580 to \$850 over the first 9 periods beyond which the recommended price gradually decreases to \$745 by period 25. In contrast, the GBM suggests a modest initial increase in price from \$610 to \$640 over the first 4 periods beyond which the recommended price falls steeply to \$200 by period 25. This results in 13.8 % lower lifetime revenues than those yielded by the optimal price path determined using the *BLDM*. This exercise reiterates the importance of having an accurate estimate of price on adoption in deriving normative pricing prescriptions for new products.

⁸ The results of these numerical simulations are reported in our Web Appendix F.

[Insert Figure 3 here]

Next we study the impact of p and q on the optimal pricing policy. We keep M and β at the average of their estimated values across the four categories (30,880 and -0.005, respectively). Based on the range of estimated values of p and q across the four categories, we choose the following three possible values for p : 0.01 (low), 0.04 (medium), 0.07 (high), and the following three possible values for q : 0.4 (low), 0.7 (medium), 1.0 (high). This yields nine possible combinations of p and q . We present the implied optimal pricing policies for these nine conditions in Figure 4. We see that at low values of q , the optimal pricing policy shows a price increase for several (10-15) periods after launch after which price flattens out and remains roughly constant over time. At medium and high values of q , the optimal pricing policy is non-monotonic, i.e., initially increasing then decreasing. At medium values of q , the launch price is higher than that at low values of q , and the subsequent price increase is also steeper, but the increase does not last for long. Between 7 and 12 periods after launch, the price reaches its peak and starts decreasing. These effects get even more pronounced at high values of q , i.e., higher launch price, steeper subsequent price increase, and faster subsequent turnaround. Holding p constant, changing q from low (medium) to medium (high) shifts the optimal price policy upward in a more pronounced manner than changing p from low (medium) to medium (high) at a constant q . As long as q is high, the eventual price path is downward sloping. However, a decreasing optimal pricing policy from the time of launch is never found to be optimal under any combination of p and q . The reason for this, which we uncover using numerical simulations, is that we assume that revenue maximization governs the firm's pricing objective (since we do not know the firm's cost function). On the other hand, as discussed earlier, if profit maximization is the firm's objective and the firm has a steep experiential learning curve for the marginal cost of production for its new product, we uncover declining optimal pricing policies (please see our Web Appendix F for the details of this exercise). Our dynamic optimization method can still be applied under such a scenario. *Our main contribution is to provide an easy-to-use prescriptive analytics template, as described in the previous section, where a dynamically optimal pricing policy can be numerically derived by a manager for their new product using*

the utility-based extension of the BM, as long as the manager has estimates of (p, q, M, β) , for their new product based on either historical data or product analogs and surveys.

[Insert Figure 4 here]

Next, we study the impact of the price sensitivity (β) on the optimal pricing policy. We keep p and q at the average of their estimated values across the four categories (0.03 and 0.65, respectively). Based on the estimated values of β across the four categories, we let β vary from a low value of -0.001 to a high value of -0.01. We present the implied introductory prices in Figure 5 (the optimal price paths are non-monotonic, i.e., first increasing and then decreasing,). We see a steep decline in the implied introductory price as the market becomes more price sensitive (varying from \$3500 to \$500). Beyond a threshold level of price sensitivity (-0.003 or so), the decrease in optimal introductory price almost flattens out. *It is useful to note here that managers can explicitly derive optimal introductory prices for their new products, as illustrated here, based on the utility-based extension of the BM, something that is impossible using the GBM (where only the post-launch price path, conditional on an assumed introductory price, can be derived from the model).*

[Insert Figure 5 here]

Next, we study the impact of the firm's time discount factor (ρ) on the optimal pricing policy. We let ρ vary from 0 to 1. We present the implied introductory prices in Figure 6 (the optimal price paths are non-monotonic, i.e., first increasing and then decreasing,). We see a very slow increase in the implied introductory price as the discount factor increases from 0 (myopic) to about 0.7, and then a steep increase thereafter. For example, an increase of the discount factor from 0.8 to 0.9 implies an increase in the introductory price from about \$250 to \$325. We present Figures 5 and 6 mainly to demonstrate that the optimal launch price, a key strategic lever in a new product manager's list of responsibilities, can be calculated using the utility-based extension of the BM for any combination of market and firm parameters. The GBM cannot be used for this purpose. The GBM requires an *assumed* launch price even to compute an optimal post-launch price path.

[Insert Figure 6 here]

Extension of The Dynamic Pricing Approach to Handle Multiple Products

In some product categories, multiple competitors launch new products at the same time, in which case consumers must decide which of multiple (if any) products to adopt at any point of time. One can handle this by extending the derived random utility model (*BGDM* / *BLDM*) to include all available products, in addition to the no purchase option, as choice alternatives for consumers. Let us consider, for example, the two product case. Under the *BLDM*, this yields the following choice probabilities, Pr_{1t} , Pr_{2t} and Pr_{0t} , for consumer adoption for either of the two available products, or the no-purchase option, at time t .

$$\begin{aligned} Pr_{1t} &= \frac{e^{W_{1t}}}{e^{W_{0t}} + e^{W_{1t}} + e^{W_{2t}}}, \\ Pr_{2t} &= \frac{e^{W_{2t}}}{e^{W_{0t}} + e^{W_{1t}} + e^{W_{2t}}}, \\ Pr_{0t} &= \frac{e^{W_{0t}}}{e^{W_{0t}} + e^{W_{1t}} + e^{W_{2t}}}, \end{aligned} \quad (23)$$

where W_{1t} , W_{2t} and W_{0t} are, in turn, represented by the following equations.

$$\begin{aligned} W_{1t} &= \lambda_1 + \text{Price}_{1t} * \beta, \\ W_{2t} &= \lambda_2 + \text{Price}_{2t} * \beta, \\ W_{0t} &= -\ln \left[\ln \left\{ \frac{1 - F(t-1)}{1 - F(t)} \right\} \right], \end{aligned} \quad (24)$$

In other words, this extended version of *BLDM* embeds the category-specific diffusion component, $\ln[\ln\{[1 - F(t-1)]/[1 - F(t)]\}]$, within the no-purchase option's deterministic utility, W_{0t} , while allowing consumers' intrinsic preferences, λ_1 and λ_2 , to be asymmetric across the two products. The consumer's unconditional likelihood of adopting each product at time t is as follows.

$$\begin{aligned} L_{1t} &= \left[\prod_{u=1}^{t-1} Pr_{0u} \right] * Pr_{1t}, \\ L_{2t} &= \left[\prod_{u=1}^{t-1} Pr_{0u} \right] * Pr_{2t}, \end{aligned} \quad (25)$$

and the predicted sales of each new product in period t is, therefore,

$$\begin{aligned}\hat{N}_{1t} &= M * L_{1t}, \\ \hat{N}_{2t} &= M * L_{2t}.\end{aligned}\tag{26}$$

Using data on consumer adoption in a category with multiple products, one can use the non-linear least squares procedure, first proposed by Srinivasan and Mason (1986), to estimate model parameters. We would like to note that this extended version of the *BLDM* differs from the framework of Krishnan, Seetharaman and Vakratsas (2012) since, in their application, they used a first-stage *BLDM* to model the category adoption, and a second-stage MNL to model product shares.

In our case, to extend our dynamic pricing approach to handle the multiple product case, we must solve for the Markov-Perfect Nash Equilibrium (MPNE) of prices for the two competing products, by using a modified Pakes and McGuire (1994) algorithm that utilizes the dynamic optimization procedure developed in Cosguner, Chan and Seetharaman (2018) (see Web Appendix G for details).

In order to numerically illustrate the application of this extended dynamic pricing approach, we use the Color TV category. First, we fix $M = 52274$ (the estimated value from Table 3(a)). Second, we assume the existence of two competing products, both of whose prices are assumed to be identical to the category prices observed in our Color TV dataset. Third, we simulate product-level adoptions data from the observed category-level adoptions in our Color TV dataset by varying λ_1 and λ_2 such that the relative baseline market share of product 1, relative to product 2, takes four possible values: 20 %, 40 %, 60 % and 80 %. This results in four synthetic product-level adoption datasets. Fourth, we estimate the parameters of the extended BLDM -- $p, q, \beta, \lambda_1, \lambda_2$ -- by applying the non-linear least squares procedure to each of the four synthetic product-level datasets. We report the results of our estimation in Table 6. As expected, we recover identical estimates of $p = 0.007, q = 0.885, \beta = -0.005$ under all 4 datasets (since these parameters require only category-level adoption outcomes, which are identical across all 4 product-level datasets, for identification). Further, the estimates of λ_1 and λ_2 adjust in order to reflect the differences in relative baseline market shares across the two products in the four synthetic datasets.

[Insert Table 6 here]

Fifth, we implement the dynamic pricing approach explained in Web Appendix G on the four synthetic product-level datasets. The optimal price paths for the four product-level baseline market share situations are reported in Figure 7. The shapes of the optimal price paths are similar across the four situations, increasing at a diminishing rate over time and saturating in 10 periods or so, with the optimal price level increasing with relative baseline market share, as expected. For example, the saturation price varies from \$300 (for a 20 % baseline share product) to \$475 (for an 80 % baseline share product).

[Insert Figure 7 here]

Last, we implement the dynamic pricing approach for a situation when both products are owned by a common (monopolist) manufacturer. We assume consumers' intrinsic preferences to be identical across the two products, i.e., $\lambda_1 = \lambda_2$. The optimal price path yielded by this analysis is reported in Figure 8. For comparison, we also plot the optimal price path for the case when the two products are owned by competitors (as in Figure 7). As expected, we find that the optimal prices are always higher in the common ownership case. For example, the saturation price is \$660 in the common ownership case versus \$380 in the competing owners case. We also find that the time path of prices shows a smaller increase over time in the competing owners case as opposed to the common ownership case.

[Insert Figure 8 here]

Conclusions

The Bass Model (BM) has well served the managerial purpose of new product sales forecasting in thousands of industrial applications over the past few decades. While pre-launch sales forecasting is important for new product planning, an equally important responsibility for a product manager is the design of optimal marketing policies, especially launch price and advertising, for the new product. In this regard, the most successful sales forecasting model for new products, the Generalized Bass Model (GBM), is found to be severely lacking. This is because the GBM incorporates marketing mix effects within the BM using percentage changes in marketing variables from period to period, and not using their absolute levels themselves. Therefore, it is necessary for the manager to make additional, untested, assumptions before

rendering the GBM usable for optimal price policy planning. Even after making such untested assumptions, the GBM can be used to derive only the optimal price path or advertising path for the new product after launch, conditional on an assumed launch price, but not the launch price or advertising level itself. One solution to this problem is to estimate pre-launch consumer price sensitivity using a conjoint survey, and then derive the launch price based on this price sensitivity. However, given the utility foundation of a conjoint model, such a launch price may not be behaviorally consistent with the post-launch price path yielded by the non-utility based GBM.

In this paper, we provide a utility-based extension of the BM (unlike the GBM) that does not suffer from the prescriptive weakness of the GBM. Unlike other utility-based diffusion models in the literature, the provided utility-based diffusion model nests the BM as a special case. Built on the foundations of consumer utility maximization, the diffusion model allows the adoption rate for the new product to directly depend on the absolute levels of the marketing mix. Therefore, the diffusion model can yield, for example, normative prescriptions regarding both the introductory price, as well as the optimal price path after launch, for the new product. We offer two versions of the utility-based diffusion model, namely the *Bass-Gumbel Diffusion Model (BGDM)* and the *Bass-Logit Diffusion Model (BLDM)*, the latter of which has been previously used by Krishnan, Seetharaman and Vakratsas (2012). The *BGDM* version that is new to the literature nests the BM as a special case. Given its utility foundation, one can estimate *BGDM / BLDM* in the absence of past sales data, using choice-based conjoint data instead.

Using empirical data from 4 product categories, we show that the two versions of the utility-based diffusion model handily outperform the GBM and BM in forecasting new product sales both in sample and out of sample, with the *BLDM* outperforming the *BGDM*. On the basis of these findings, we propose that even if one's purpose is to only forecast new product sales, and not optimize the new product's marketing mix, the *BGDM / BLDM* are still to be preferred to the GBM. *BGDM / BLDM* also predict the peak time of adoptions more accurately than the GBM. By using a computationally convenient dynamic pricing approach, we explicitly compare the optimal pricing policy yielded by the *BLDM* to that of the GBM and find that the optimal price path derived from GBM yields 13.8% lower revenue than the price path derived

from *BLDM*. We also derive the optimal pricing policies under a range of possible values of model parameters. Finally, we show how to extend the dynamic pricing approach to the competitive, multi-product context.

Some important caveats are in order. First, using adoption data across many new products, it would be interesting to study the implications of varying degrees of heterogeneity in price sensitivities across consumer segments on the implied optimal price paths for the product.⁹ Second, while estimating the utility-based diffusion model using historical sales data, we treat prices as exogenous. In reality, new product prices are chosen by product managers for endogenous reasons that are not observed by the researcher. One could use a control function approach to correct for possible endogeneity of prices (if suitable instruments are available) by including the fitted residuals from an instrumental variable regression as an additional explanatory variable in the consumer's utility function. Third, we treat consumers as myopic in their decision-making, i.e., they do not anticipate the trajectory of future prices. However, there is a literature in economics and marketing, inspired by Coase (1972), that has studied the pricing implications of forward-looking consumers for durable goods sellers. For example, Besanko and Winston (1990) and Nair (2007) show that the maximum price that a monopolist will charge over the dynamic price path would be lower, and subsequent prices would decline less steeply, when consumers are forward looking. It would be interesting to study the implications of forward-looking consumers in our case to see whether the motivation to eventually decrease price would be similarly mitigated for the seller. We leave this for future research.

Our focus in this study is to provide a utility-based extension of the BM which not only forecasts new product sales better than existing models such as GBM, but also has excellent prescriptive properties for optimal price planning, and explicitly undertaking endogeneity correction is not germane to this objective. The provided dynamic pricing algorithm will yield practical value to new product managers. As a fruitful avenue for future research, we propose the study of functional regression, first proposed by Sood,

⁹ In our dataset, we estimated a two-segment model and found the optimal prices to slightly increase, while maintaining the same non-monotonic shape over time. We thank an anonymous reviewer for encouraging us to study this issue.

James and Tellis (2009), which exploits data across multiple products and markets, as an improved estimation procedure for the provided utility-based diffusion model.

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TABLE 1: FIT COMPARISON OF DIFFUSION MODELS – FULL SAMPLE

SSE	BGDM	BLDM	GBM	BM	T
Color TV	457,051	457,387	402,514	996,421	10
Air Conditioner	90,726	89,550	294,659	341,468	13
Clothes Dryer	129,485	129,227	210,770	212,777	13
Freezer	136,767	136,728	209,251	242,672	15

T = Number of Observations

Note: The best-fitting model for each product category is shown in bold.

TABLE 2: PREDICTIONS COMPARISON – HOLDOUT VALIDATION

SSE	Actual	BGDM	BLDM	GBM	BM	Period
Color TV	5962	5506	5524	4705	4288	9
	4631	4951	4977	3102	2527	10
AC	1800	1688	1695	1244	1283	11
	1580	1453	1471	862	947	12
	1500	1466	1488	648	652	13
Dryer	1425	1041	1064	942	991	11
	1260	841	878	631	739	12
	1236	635	684	472	521	13
Freezer	1205	988	989	866	829	11
	1045	988	988	754	724	12
	1050	905	910	620	620	13

Note: The best-fitting model for each product category / period combination is shown in bold.

TABLE 3: PREDICTION COMPARISON: K-FOLD CROSS VALIDATION
(Entries represent the % of times each model is the “best” across all folds)

	Leave One Out			Leave Two Out		
	BGDM	BLDM	GBM	BGDM	BLDM	GBM
Models with Price						
Color TV	30%	30%	40%	24%	38%	38%
Air Conditioner	46%	23%	31%	32%	37%	31%
Clothes Dryer	54%	15%	31%	32%	40%	28%
Freezer	47%	13%	40%	36%	26%	38%

TABLE 4 (a): PARAMETER ESTIMATES FOR THE FOUR DIFFUSION MODELS

	Parameter	BGDM	BLDM	GBM	BM
Color TV	p	0.007	0.006	0.004	0.005
	q	0.887	0.890	0.591	0.635
	M	52,290	52,274	40,273	40,248
	β	-0.003	-0.003	-5.026	na
Air Conditioner	p	0.030	0.029	0.009	0.009
	q	0.800	0.798	0.361	0.375
	M	27,848	28,542	18,820	18,710
	β	-0.008	-0.008	-0.812*	na

Clothes Dryer	p	0.030	0.029	0.014	0.014
	q	0.455	0.461	0.327	0.327
	M	20,350	20,802	16,387	16,497
	β	-0.006	-0.006	-0.302*	na
Freezer	p	0.058	0.057	0.020	0.025
	q	0.435	0.439	0.143	0.148
	M	20,932	21,437	25,578	22,266
	β	-0.004	-0.004	1.301*	na

Note: * denotes that the coefficient is not significant at the 0.05 level.

TABLE 4 (b): MARKET POTENTIAL ESTIMATES USING TRAINING SAMPLE

	BGDM	BLDM	GBM	BM
Color TV	59,961	61,317	33,729	31,608
Air Conditioner	25,720	26,845	13,467	13,660
Clothes Dryer	12,683	13,183	11,532	11,844
Freezer	18,971	19,414	16,961	15,774

Note: The estimates in this table are generated using only the first 80 % of the observations

TABLE 5: PREDICTED PEAK ADOPTION TIMES AND NUMBER OF ADOPTIONS

		BGDM	BLDM	BM	GBM	Actual
Color TV	Peak Adoption Time	8	8	8	8	8
	Peak Number of Adoptions	6,139	6,140	6,442	6,218	5,982
	Adoption Lifetime	37	37	20	20	NA
Air Conditioner	Peak Adoption Time	8	8	10	10	8
	Peak Number of Adoptions	1,784	1,778	1,837	1,933	1,828
	Adoption Lifetime	48	48	27	27	NA
Clothes Dryer	Peak Adoption Time	8	8	10	10	8
	Peak Number of Adoptions	1,412	1,409	1,457	1,454	1,523
	Adoption Lifetime	45	47	28	28	NA
Freezer	Peak Adoption Time	14	14	11	9	13
	Peak Number of Adoptions	1,108	1,106	1,122	1,117	1,205
	Adoption Lifetime	46	50	47	49	NA

TABLE 6: BLDM ESTIMATES WITH TWO ASYMMETRIC PRODUCTS

Market Share of Product 1:	20%	40%	60%	80%
p	0.007	0.007	0.007	0.007
q	0.885	0.885	0.885	0.885
M	52274	52274	52274	52274
λ_1	-0.413	0.277	0.682	0.972
λ_2	0.972	0.682	0.277	-0.413
B	-0.005	-0.005	-0.005	-0.005

Note: All estimates are significant at the 5 % level. M is set to 52274 in the estimation for identification purpose.

FIGURE 1: ESTIMATED PROBABILITY DENSITIES OF ADOPTION TIMES

Figure 1A: Color TV

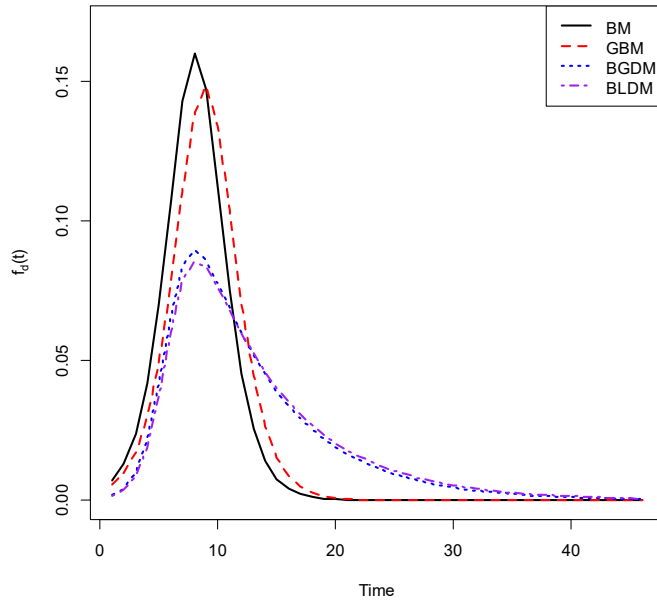


Figure 1B: Air Conditioner

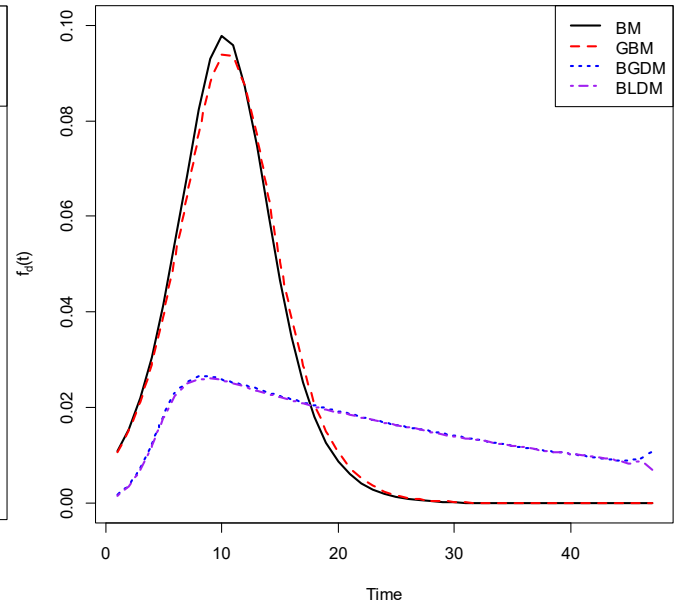


Figure 1C: Clothes Dryers

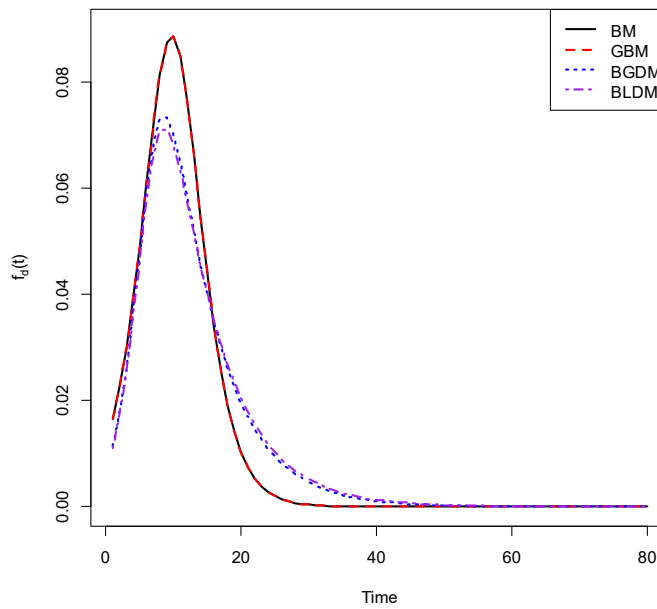


Figure 1D: Freezers

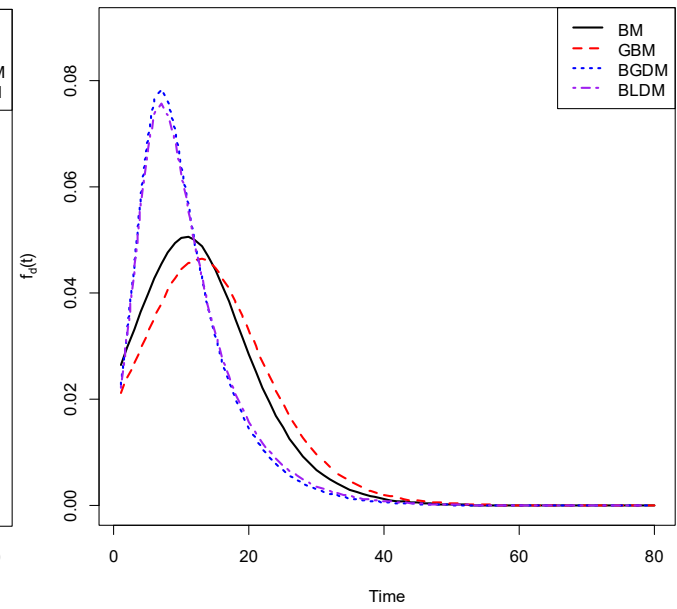


FIGURE 2: OPTIMAL DYNAMIC PRICE PATHS YIELDED BY THE BLDM

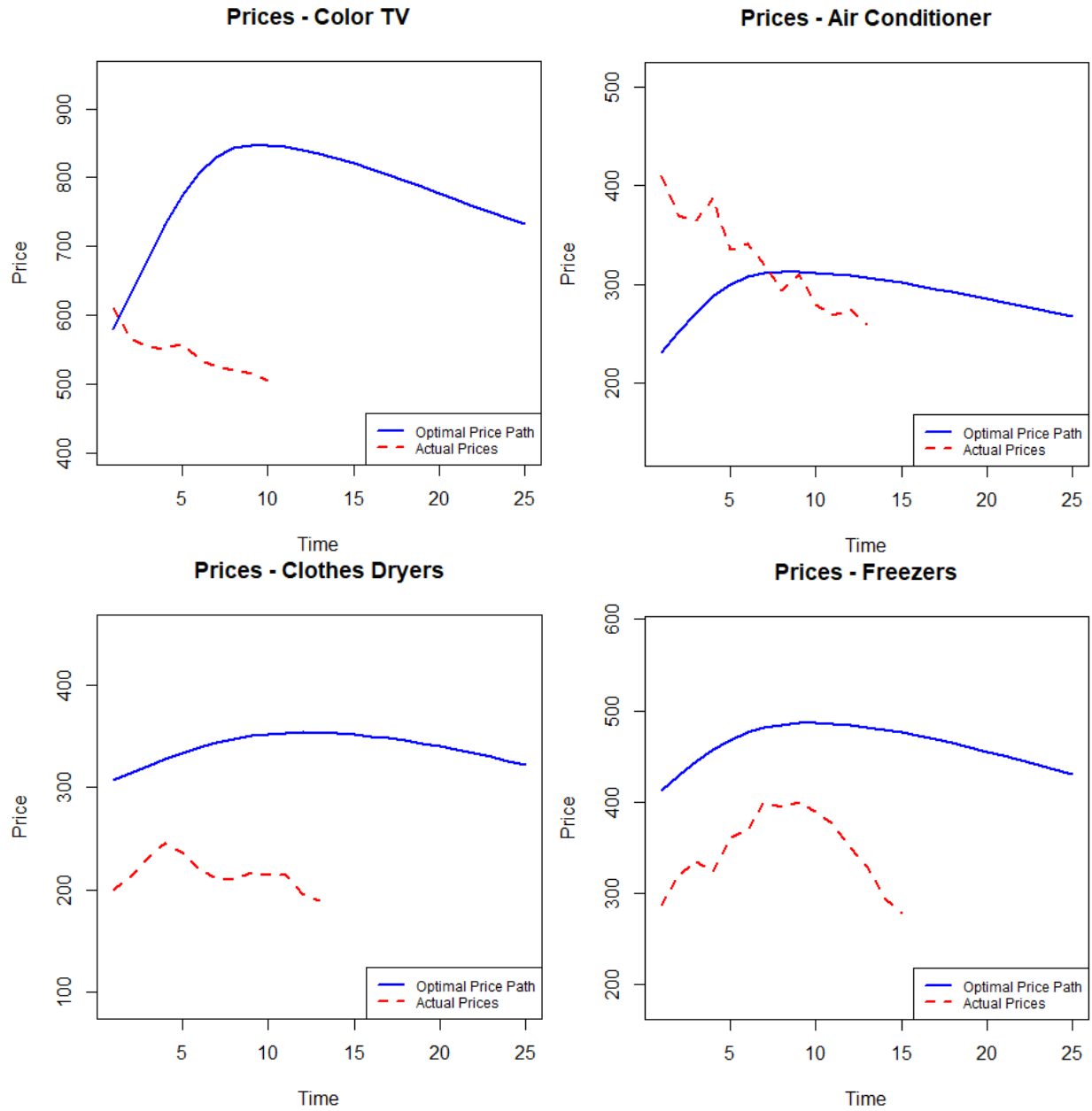


FIGURE 3: OPTIMAL DYNAMIC PRICE PATH OF BLDM VERSUS GBM: COLOR TV

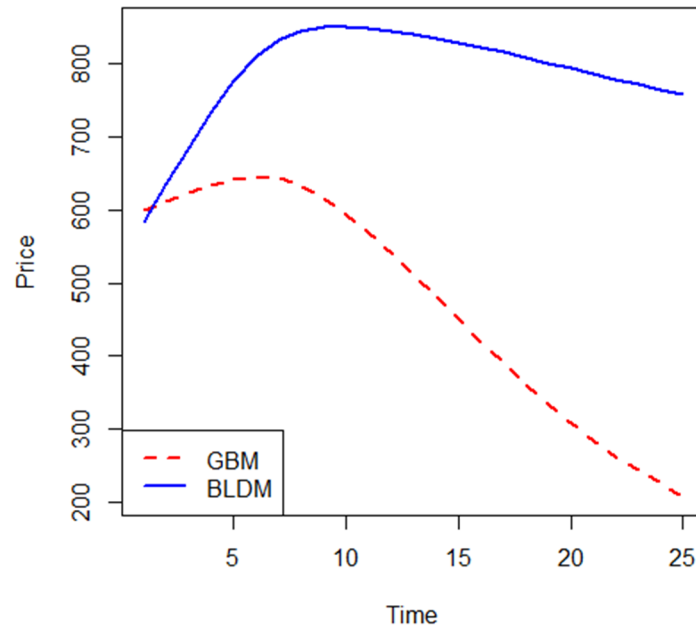


FIGURE 4: OPTIMAL DYNAMIC PRICING POLICY IMPLIED BY THE BLDM

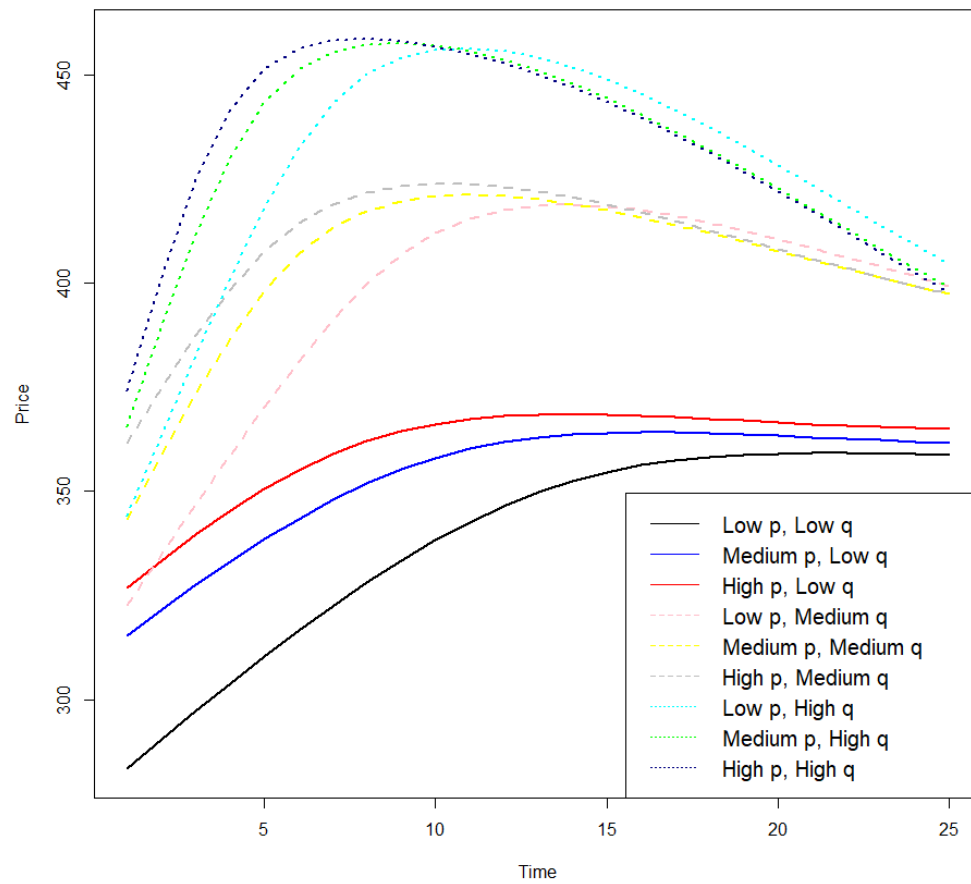


FIGURE 5: OPTIMAL INTRODUCTORY PRICE AT VARIOUS LEVELS OF MARKET PRICE SENSITIVITY (β)

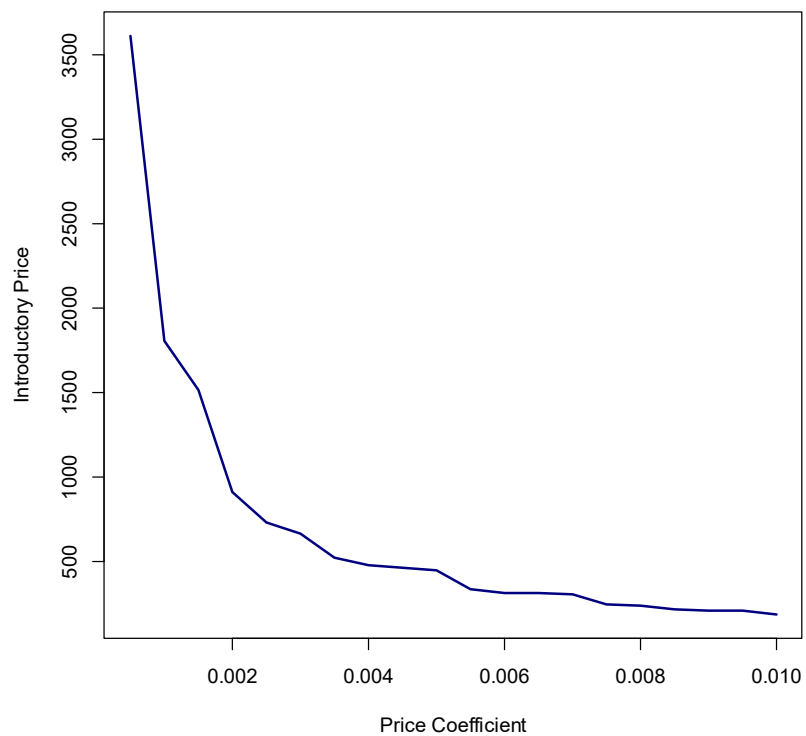


FIGURE 6: OPTIMAL INTRODUCTORY PRICE AT VARIOUS LEVELS OF FIRM DISCOUNT FACTOR (ρ)

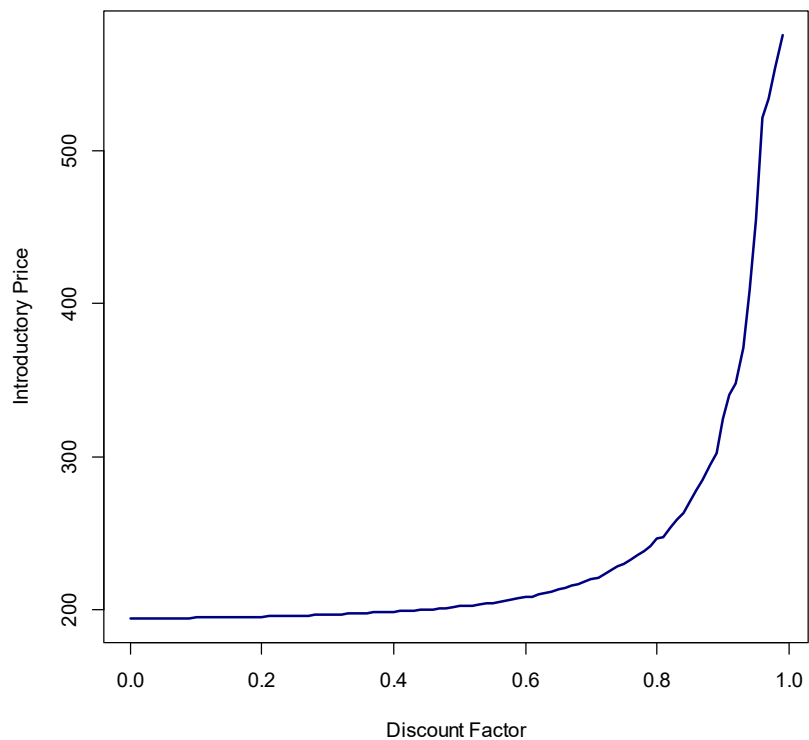


FIGURE 7: OPTIMAL DYNAMIC PRICE PATHS OF BLDM – TWO COMPETING PRODUCTS

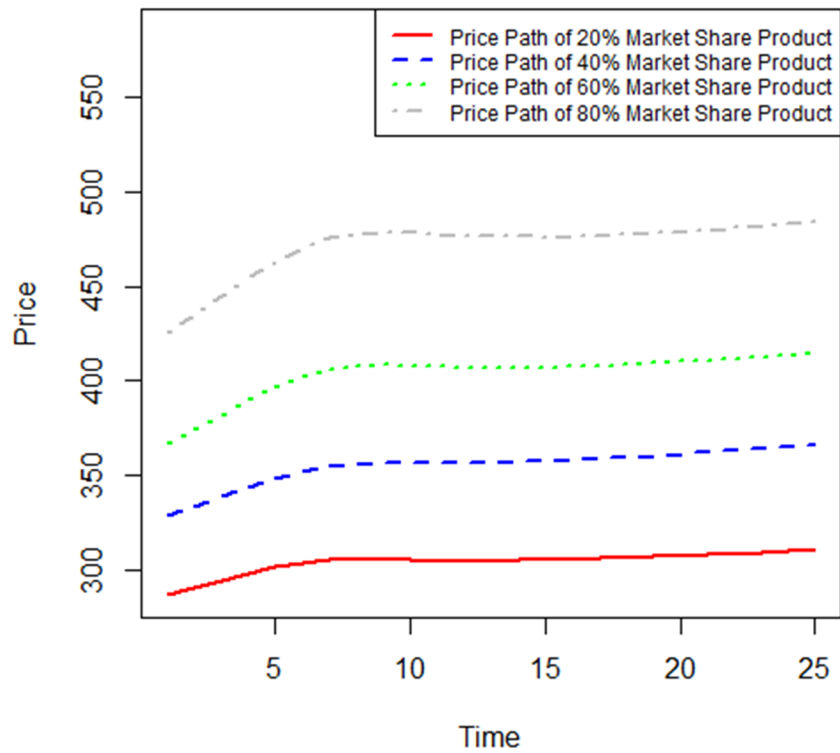
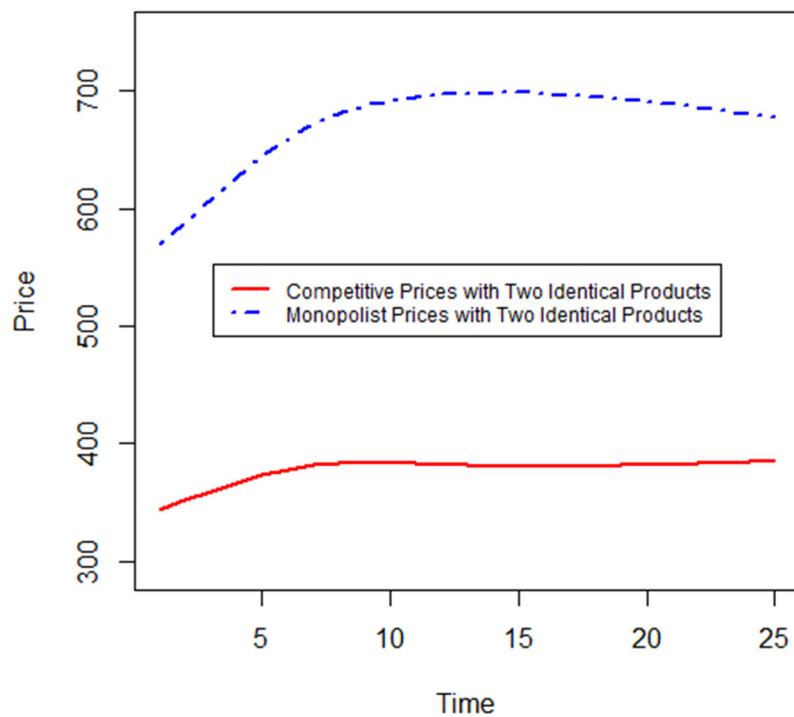


FIGURE 8: OPTIMAL DYNAMIC PRICE PATH OF BLDM – MONOPOLIST SELLING TWO PRODUCTS



WEB APPENDICES

Web Appendix A: Holdout Validation

We construct the holdout sample in 4 different ways: (i) final 10 % of observations, (ii) final 20 % of observations (reported in the main manuscript), (iii) final 30 % of observations, (iv) final 40 % of observations. We do not use cases (iii) and (iv) for Color TV because of the relative paucity of observations in that category. Furthermore, we do not use additional cases, such as having more than 40 % of observations in the holdout sample since the predictions become unreliable, as is well known in the diffusion modeling literature, with such small training samples, especially when the data span a period sufficiently prior to the projected peak times of adoption. In practice, product analogs play an important role in such early stages of the adoption cycle, where the implied adoption rates of the analogs are combined with limited historical data on the product, using a Bayesian updating scheme (see Lenk and Rao 1990), in order to make reliable forecasts. Since we are only using historical data in our study and ignoring product analogs, we study only the 4 cases listed above. In the following table, we present the results of holdout predictions in all 4 cases for each holdout observation.

Scenario (i) – 90% Training Sample						
	Actual	BGDM	BLDM	GBM	BM	Periods
Color TV	4631	5776	5774	4539	4119	10
AC	1500	1614	1619	1279	1221	13
Dryer	1236	1176	1180	951	977	13
Freezer	1050	1032	1037	1071	941	15

Scenario (ii) – 80% Training Sample						
	Actual	BGDM	BLDM	GBM	BM	Periods
Color TV	5962	5506	5524	4705	4288	9
	4631	4951	4977	3102	2527	10
AC	1800	1688	1695	1244	1283	11
	1580	1453	1471	862	947	12
	1500	1466	1488	648	652	13
Dryer	1425	1041	1064	942	991	11
	1260	841	878	631	739	12
	1236	635	684	472	521	13
Freezer	1205	988	989	866	829	13
	1045	988	988	754	724	14
	1050	905	910	704	620	15

Scenario (iii) – 70% Training Sample

	Actual	BGDM	BLDM	GBM	BM	Periods
AC	1673	1955	1953	1681	1332	10
	1800	2031	2030	1252	970	11
	1580	1796	1803	869	643	12
	1500	1967	1973	655	401	13
Dryer	1240	1289	1327	928	1211	10
	1425	1127	1207	636	966	11
	1260	991	1141	321	713	12
	1236	805	992	247	497	13
Freezer	925	903	905	834	821	11
	1101	824	827	702	684	12
	1205	727	734	587	552	13
	1045	644	654	465	434	14
	1050	540	555	384	335	15

Scenario (iv) – 60% Training Sample						
	Actual	Gumbel	Logit	GBM	Bass	Periods
AC	1586	1441	1448	1341	1784	9
	1673	1865	1873	1449	1596	10
	1800	1925	1938	963	1265	11
	1580	1688	1709	608	909	12
	1500	1841	1867	450	609	13
Dryer	1294	1835	1839	1762	1772	9
	1240	1986	2004	1528	1900	10
	1425	2000	2044	1321	1904	11
	1260	1792	1876	859	1782	12
	1236	1531	1653	756	1565	13
Freezer	975	1002	1005	914	911	10
	925	936	941	787	772	11
	1101	870	878	649	629	12
	1205	782	795	532	497	13
	1045	709	725	414	383	14
	1050	605	626	336	290	15

We summarize the key take-aways of the above table below:

1. In case (i), there are 4 holdout observations in total. *BLDM* wins in 2 cases (50%), *BGDM* in 1 case (25%), and *GBM* in 1 case (25%). Overall, therefore, our utility-based models win in 3 out of 4 cases (75%).
2. In case (ii), there are 11 holdout observations in total. *BLDM* wins in 10 cases (91%), *BGDM* in 1 case (9%). Overall, therefore, our utility-based models win in 11 out of 11 cases (100%).

3. In case (iii), there are 13 holdout observations in total. *BLDM* wins in 9 cases (69%), *BGDM* in 3 cases (23%), and GBM in 1 case (8%). Overall, therefore, our utility-based models win in 12 out of 13 cases (92%).
4. In case (iv), there are 16 holdout observations in total. *BLDM* wins in 5 cases (31%), *BGDM* in 7 cases (44%), and GBM in 4 cases (25%). Overall, therefore, our utility-based models win in 12 out of 16 cases (75%).

We also present a convenient qualitative summary of these analyses in the table below.

Training Sample Size	BGDM	BLDM	GBM	BM
60%	44%	31%	25.0%	0.0%
70%	23%	69%	8%	0.0%
80%	9%	91%	0.0%	0.0%
90%	25%	50%	25%	0.0%

These analyses lend additional support to the predictive ability of *BGDM* / *BLDM*.

Web Appendix B.1: Prior Models of New Product Adoption with Marketing Mix Variables

We compare the empirical performance of the *BLDM* against that of each of several previously proposed diffusion models which incorporate the effects of marketing variables. These models are listed below.

Robinson and Lakhani (1975)

$$\rightarrow Sales_t = (M - [\sum_{k=1}^{t-1} Sales_k])(p + q [\sum_{k=1}^{t-1} Sales_k])e^{-\beta Price_t}$$

Horsky and Simon (1983)

$$\rightarrow Sales_t = (M - [\sum_{k=1}^{t-1} Sales_k])(\alpha + \beta \ln(Price_t) + q [\sum_{k=1}^{t-1} Sales_k])$$

Simon and Sebastian (1987)

$$\rightarrow \text{Specification A: } Sales_t = (M - [\sum_{k=1}^{t-1} Sales_k])(p + p f(Price_t) q [\sum_{k=1}^{t-1} Sales_k])$$

$$\rightarrow \text{Specification B: } Sales_t = (M - [\sum_{k=1}^{t-1} Sales_k])(p + \{q + q f(Price_t)\} [\sum_{k=1}^{t-1} Sales_k])$$

where $f(Price_t)$ is either $\ln(Price_t)$ or $\sum_{k=1}^t \ln(Price_k)$. We call these two cases A1 (B1) and A2 (B2), respectively.

Jain and Rao (1990)

→ Specification A: $Sales_t = (M Price_t^{-\beta} - [\sum_{k=1}^{t-1} Sales_k]) \frac{F(t)-F(t-1)}{1-F(t-1)}$

→ Specification B: $Sales_t = (M - [\sum_{k=1}^{t-1} Sales_k])(Price_t^{-\beta}) \frac{F(t)-F(t-1)}{1-F(t-1)}$

where $F(t) = \frac{(1 - e^{-(p+q)t})}{(1 + (q/p)e^{-(p+q)t})}$

In the following table, we report the minimized criterion function (sum of squared errors) associated with each model estimated on our four datasets.

Model	Color TV	AC	Dryer	Freezer
BLDM	457,387	89,550	129,227	136,728
Robinson and Lakhani (1975)	2,052,148	535,245	249,838	223,199
Horsky and Simon (1983)	2,184,305	248,158	266,154	251,550
Simon and Sebastian (1987) – A1	2,742,153	542,835	269,617	257,846
Simon and Sebastian (1987) – A2	1,744,018	310,979	170,260	375,005
Simon and Sebastian (1987) – B1	2,702,510	530,939	278,377	269,620
Simon and Sebastian (1987) – B2	5,641,234	924,557	523,261	409,141
Jain and Rao (1990) – A	573,609	226,088	130,852	135,236
Jain and Rao (1990) – B	427,858	144,657	128,350	143,110

Except the Jain and Rao (1990) models, the remaining models are vastly outperformed by the *BLDM*. However, the Jain and Rao (1990) – B model outperforms our model in the Color TV category, and narrowly beats it in the dryer category. While we acknowledge the good predictive performance of the Jain and Rao (1990) – B model, *we must note that it cannot be used for price optimization since it implies positive infinity as the profit-maximizing price*. On the other hand, our model can be used for price optimization, which is the focus of our research.

We report the parameter estimates yielded by the Jain and Rao (1990) models below.

	Parameter	BLDM	Jain & Rao - A	Jain & Rao - B
Color TV	p	0.006	0.006	0.004
	q	0.890	0.668	0.887
	M	52,720	unstable *	53,410
	β	-0.0025	-3.05 *	-0.178
Air Conditioner	p	0.029	0.014	0.013
	q	0.803	0.381	0.768
	M	28,560	unstable *	46,860
	β	-0.008	-1.095 *	-0.444

Clothes Dryer	p	0.029	0.014	0.026
	q	0.461	0.377	0.498
	M	20,800	unstable *	25,610
	β	-0.006	-0.826 *	-0.296 *
Freezer	p	0.057	0.024	0.098
	q	0.439	0.310	0.432
	M	21,440	unstable *	44,060
	β	-0.004	-0.802	-0.440

Note: * denotes that the coefficient is not significant at the 0.05 level.

Jain and Rao (1990) – A does not yield stable or statistically significant estimates of market potential (M). The estimates of M are generally implausibly large, with correspondingly large standard errors (the *BLDM* does not suffer from this concern). The price coefficients, while negatively signed, are statistically insignificant in 3 out of the 4 categories (while the *BLDM* recovers statistically significant price coefficients in all categories). On the other hand, Jain and Rao (1990) – B yields more sensible parameter estimates (although the estimates of M are overstated by a factor of 2, compared to our model, in two categories). That said, as we have noted earlier, Jain and Rao (1990) - B is limited from the price optimization perspective since it implies positive infinity for the level of optimal prices.

Web Appendix B.2: Proportional Hazard Models (PHM) of New Product Adoption

This online appendix compares the *BLDM*'s empirical performance versus various proportional hazard models (PHM) with marketing variables. Specifically, we test five different specifications of the PHM (each using a different baseline hazard function) as listed below.

Exponential Hazard: $h(t) = \gamma$

$$\rightarrow \Pr(\text{Purchase}) = 1 - e^{-e^{\beta \text{Price}} * \gamma}$$

Erlang-2 Hazard: $h(t) = (\gamma^2 t) / (1 + \gamma t)$

$$\rightarrow \Pr(\text{Purchase}) = 1 - e^{-e^{\beta \text{Price}} * \{\gamma + \ln[1 + (t-1)\gamma] - \ln[1 + t\gamma]\}}$$

Weibull Hazard: $h(t) = \gamma \alpha (\gamma t)^{\alpha-1}$

$$\rightarrow \Pr(\text{Purchase}) = 1 - e^{-e^{\beta \text{Price}} * \{[t\gamma]^\alpha - [(t-1)\gamma]^\alpha\}}$$

Log-logistic Hazard: $h(t) = [\gamma \alpha (\gamma t)^{\alpha-1}] / [1 + (\gamma t)^\alpha]$

$$\rightarrow \Pr(\text{Purchase}) = 1 - e^{-e^{\beta \text{Price}} * \{\ln[1+(t\gamma)^\alpha] - \ln[1+((t-1)\gamma)^\alpha]\}}$$

$$\text{Expo-power Hazard: } h(t) = \gamma \alpha t^{\alpha-1} e^{\theta t^\alpha}$$

$$\rightarrow \Pr(\text{Purchase}) = 1 - e^{-e^{\beta \text{Price}} * \left\{ \frac{\gamma}{\theta} \left[e^{\theta t^\alpha} - e^{\theta (t-1)^\alpha} \right] \right\}}$$

In the following table, we report the minimized criterion function (sum of squared errors) associated with five different specifications of the PHM (each using a different baseline hazard function) estimated on our four datasets.

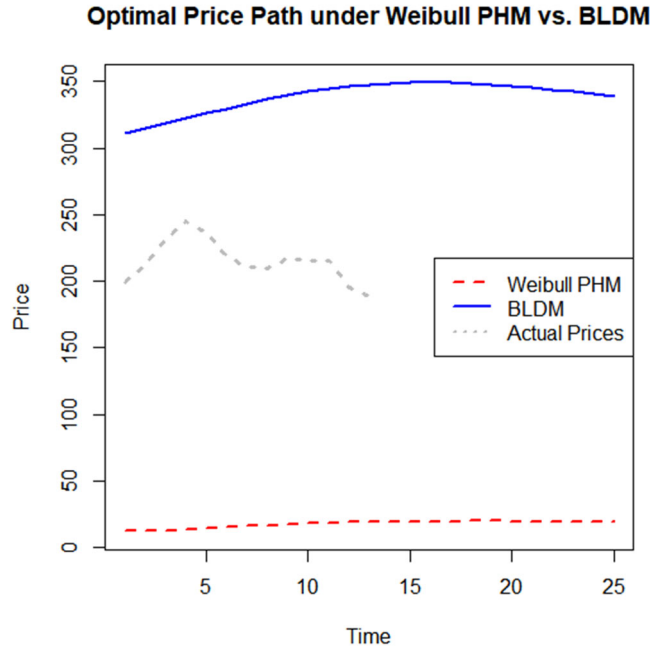
Model	Color TV	AC	Dryer	Freezer
BLDM	457,387	89,550	129,227	136,728
PHM – Exponential	4,880,824	702,022	1,180,811	623,342
PHM – Erlang2	4,880,824	573,859	293,558	221,601
PHM – Weibull	575,972	150,120	124,395	185,567
PHM – Log Logistic	618,587	98,285	155,528	185,530
PHM – Expo Power	653,067	170,205	138,736	195,639

The *BLDM* outperforms all five specifications of the PHM in 3 out of 4 categories. In the dryer category, the Weibull specification of the PHM outperforms our utility-based model. We report the parameter estimates yielded by PHM - Weibull in the dryer category below.

Parameter	BLDM	PHM - Weibull
p	0.029	na
q	0.461	na
δ	na	0.200 *
α	na	2.197
M	20,800	16,350
β	-0.006	-0.008 *

Note: * denotes that the coefficient is not significant at the 0.05 level.

We see that while the price coefficient under PHM – Weibull is negatively signed, as expected; however, it is not statistically significant. That said, we still derive the optimal price path implied by the PHM – Weibull by using the estimated size of the price coefficient and report the plot below. *The optimal prices implied by the PHM – Weibull appear to be unrealistically low and almost constant over time.* The optimal prices implied by our utility-based model, on the other hand, are much closer to the actual prices observed in the data.



Web Appendix C.1: Covid-19 Diffusion

To illustrate the empirical performance of *BGDM* / *BLDM* on recent data, we first focus on an epidemiological application. It is useful to note here that marketing diffusion models were originally inspired by mathematical models of disease spread that were developed in epidemiology. Specifically, we study the effect of social distancing (covariate) on the spread of COVID-19 (disease diffusion) in Missouri.

We located two data sources to enable the above. The first contains daily new positive cases in each of 444 school districts in Missouri and was obtained from Missouri Hospital Association. The second contains social distancing measures, collected using GPS location data from a panel of cellphone users, and was obtained from SafeGraph (<http://www.safegraph.com/covid-19-data-consortium>). We use three time-varying metrics of social distancing: (1) percentage of residents staying home, (2) percentage of residents working full-time at their workplace, (3) percentage of residents working part-time at their workplace. These three metrics serve as covariates in our epidemiological application (similar to price in our marketing applications).

We aggregate daily cases, as well as the covariates, to the weekly level and obtain 18 usable weekly observations. These observations represent the 18-week period from July 1, 2020 to October 31, 2020. We have a total of 444 datasets representing the 444 school districts. We estimate four diffusion models --BM, GBM, BGDM, and BLDM -- in each of the 444 datasets. We find that the utility-based models (either

BGDM or BLDM) yield the best fit of the observed diffusion of Covid-19 in 75.45 % of the datasets, while the GBM outperforms our utility-based models in 24.55 % of the datasets. Social distancing measures are observed to significantly influence the spread of disease in expected ways (i.e., more social distancing is associated with slower disease spread). Detailed results are available upon request.

This further vindicates the predictive performance of *BGDM* / *BLDM*, albeit in the context of epidemiology. If one were a public policy planner, one could treat the social distance metrics as variables that can be “optimized” (to arrest the fast spread of disease in the school district) by taking in to account monetary and human resource constraints faced by the school district. In principle, our dynamic pricing algorithm can be used to handle this dynamic optimization problem with a suitably specified dynamic objective.

Web Appendix C.2: Compact SUV Adoption in India

To illustrate the empirical performance of *BGDM* / *BLDM* on recent data, we focus on a new product adoption application. We obtained quarterly data on consumer adoption of compact SUVs in the Indian market starting from Q2 of 2010 until Q1 of 2015 (i.e., 20 quarters) from Otto Edge Services LLP, a marketing research company in India. In our data, we observe quarterly sales, as well as quarterly print advertising expenditures, and quarterly TV advertising expenditures, for all brands of compact SUVs in the Indian market. We aggregated the data across all brands to create a category-level dataset. We then estimated BM, GBM, *BGDM* and *BLDM* on the category-level data. In terms of minimized sum of squared errors, the fit results are as shown below:

BM:	188.8 million
GBM:	165.4 million
BGDM:	162.3 million
BLDM:	157.4 million

Again, *BGDM* / *BLDM* outperformed the GBM, with *BLDM* being most preferred. We explicitly estimate a version of *BLDM* to see whether print and TV advertising have non-linear effects on SUV sales (using quadratic terms in the utility function). Accommodating such non-linear effects of advertising is not straightforward in the GBM. We uncover non-linear advertising effects for both types of advertising. Detailed results are available upon request.

Web Appendix D: Sample R Code for the Innovator Firm’s Dynamic Price Optimization

```
# Setting the values of p, q, M, and B
```

```
p=0.01
```

```

q=0.60
M=50000
B=-0.005

# Setting the number of periods to forward simulate (NT)

NT=40

# Creating the storage matrix for prices

prices=matrix(0,NT,1)

# Setting the initial values for the pricing policy function parameters (theta)

pars=c(400,0,0,0,0,0)

# Making the cost assumption, intercept and slope are zero for the revenue optimization

c0=0
c1=0

# Creating the lifetime revenue function

revenue=function(pars){

  A=matrix(pars)

  Ntlogit=matrix(0,NT,1)
  price=matrix(0,NT,1)
  cost=matrix(0,NT,1)

  prt=matrix(0,NT,1)
  prtm=matrix(0,NT,1)

  t=1
  tl=t-1

  Ft=((1-exp(-(p+q)*t))/(1+(q/p)*exp(-(p+q)*t)))
  Ftm1=((1-exp(-(p+q)*tl))/(1+(q/p)*exp(-(p+q)*tl)))

  # We recommend to start with a first-order policy function of states. Once the estimates from the first-
  # order policy approximation are obtained, the researcher can modify the policy function by adding the
  # quadratic and interaction terms.

  X=cbind(1,t,t^2,M,M^2,t*M)

  # Calculating the first period price, sales and cost.

  price[t]=X%%A

  vt=log(log((1-Ftm1)/(1-Ft)))+B*price[t]

```

```

prt[1,]=exp(vt)/(1+exp(vt))
prtm[1,]=1-prt[1,]

Ntlogit[1]=M*prt[1]

cost[t]=c0+c1*0

# Calculating the prices, sales and costs for periods 2 to NT.

t=2
while(t<=NT){

  tl=t-1

  Ft=((1-exp(-(p+q)*t))/(1+(q/p)*exp(-(p+q)*t)))
  Ftm1=((1-exp(-(p+q)*tl))/(1+(q/p)*exp(-(p+q)*tl)))

  if(Ft<1){

    R=(M-sum(Ntlogit[1:(t-1)]))

    X=cbind(1,t,t^2,R,R^2,t*R)

    price[t]=X%%A

    vt=log(log((1-Ftm1)/(1-Ft)))+B*price[t]

    prt[t,]=exp(vt)/(1+exp(vt))
    prtm[t,]=1-prt[t,]

    Ntlogit[t]=M*prod(prtm[1:(t-1)])*prt[t]

    cost[t]=c0+c1*sum(Ntlogit[1:(t-1)])
  }

  # If the CDF of the adoption time reaches to 1, set the sales to zero.

  if(Ft==1){
    Ntlogit[t]=0
  }

  t=t+1
}

# Creating the discount vector

discount=matrix(1,NT,1)

i=1
while(i<=NT){

```

```

discount[i,]=0.95^(i-1)

i=i+1
}

# Returning the negative of lifetime revenue

return(-sum((price-cost)*Ntlogit*discount))
}

# Optimizing the lifetime revenue to find the parameters of the policy function.

pars=optim(pars,revenue,method="Nelder-Mead",control=list(maxit=500000))
pars=pars$par

# Given the estimated policy function parameters, forward simulating prices, sales and costs from period 1
to NT.

A=matrix(pars)

Ntlogit=matrix(0,NT,1)
price=matrix(0,NT,1)
cost=matrix(0,NT,1)

prt=matrix(0,NT,1)
prtm=matrix(0,NT,1)

t=1
tl=t-1

Ft=((1-exp(-(p+q)*t))/(1+(q/p)*exp(-(p+q)*t)))
Ftm1=((1-exp(-(p+q)*tl))/(1+(q/p)*exp(-(p+q)*tl)))

R=(M-sum(Ntlogit[1:(t-1)]))

X=cbind(1,t,t^2,M,M^2,t*M)

price[t]=X%%A

vt=log(log((1-Ftm1)/(1-Ft)))+B*price[t]

prt[1,]=exp(vt)/(1+exp(vt))
prtm[1,]=1-prt[1,]

Ntlogit[1]=M*prt[1]

cost[t]=c0+c1*0

t=2
while(t<=NT){

```

```

tl=t-1

Ft=((1-exp(-(p+q)*t))/(1+(q/p)*exp(-(p+q)*t)))
Ftm1=((1-exp(-(p+q)*tl))/(1+(q/p)*exp(-(p+q)*tl)))

if(Ft<1){

R=(M-sum(Ntlogit[1:(t-1)]))

X=cbind(1,t,t^2,R,R^2,t*R)

price[t]=X%%A

vt=log(log((1-Ftm1)/(1-Ft)))+B*price[t]

prt[t,]=exp(vt)/(1+exp(vt))
prtm[t,]=1-prt[t,]

Ntlogit[t]=M*prod(prtm[1:(t-1)])*prt[t]

cost[t]=c0+c1*sum(Ntlogit[1:(t-1)])
}

if(Ft==1){
  Ntlogit[t]=0
}

t=t+1
}

# Plotting the optimal price path

ranges=range(price)
plot(price,ylim=ranges,type='l',lty=2,col='blue',lwd=2,ylab="Price",xlab="Time",main="Optimal Price Path")

```

Web Appendix E: Sensitivity Analysis of Optimal Price Paths

Using the estimated standard errors of the estimated model parameters, we undertake a sensitivity analysis. Below we report the optimal price paths based on the estimated 95 % confidence intervals for p and q . We summarize two take-aways of these graphs: (1) The 95% prediction bands are quite “tight” for p , which means that the impact of uncertainty in the estimated value of p on optimal prices is minimal, (2) The 95% prediction band is wider for q , but it appears substantively wider than necessary because of our scaling of the y-axis. For example, the maximum price over the dynamic price path varies from \$770 (lower bound) versus \$880 (upper bound) in the color TV category, a difference of \$110 because of parameter

uncertainty (large but not worrisomely large). In practice, the risk aversion and objective functions of managers will determine how they make use of this uncertainty in pricing prescriptions and its effect on the category profit.

FIGURE E1: SENSITIVITY ANALYSES OF OPTIMAL PRICE PATHS WRT p

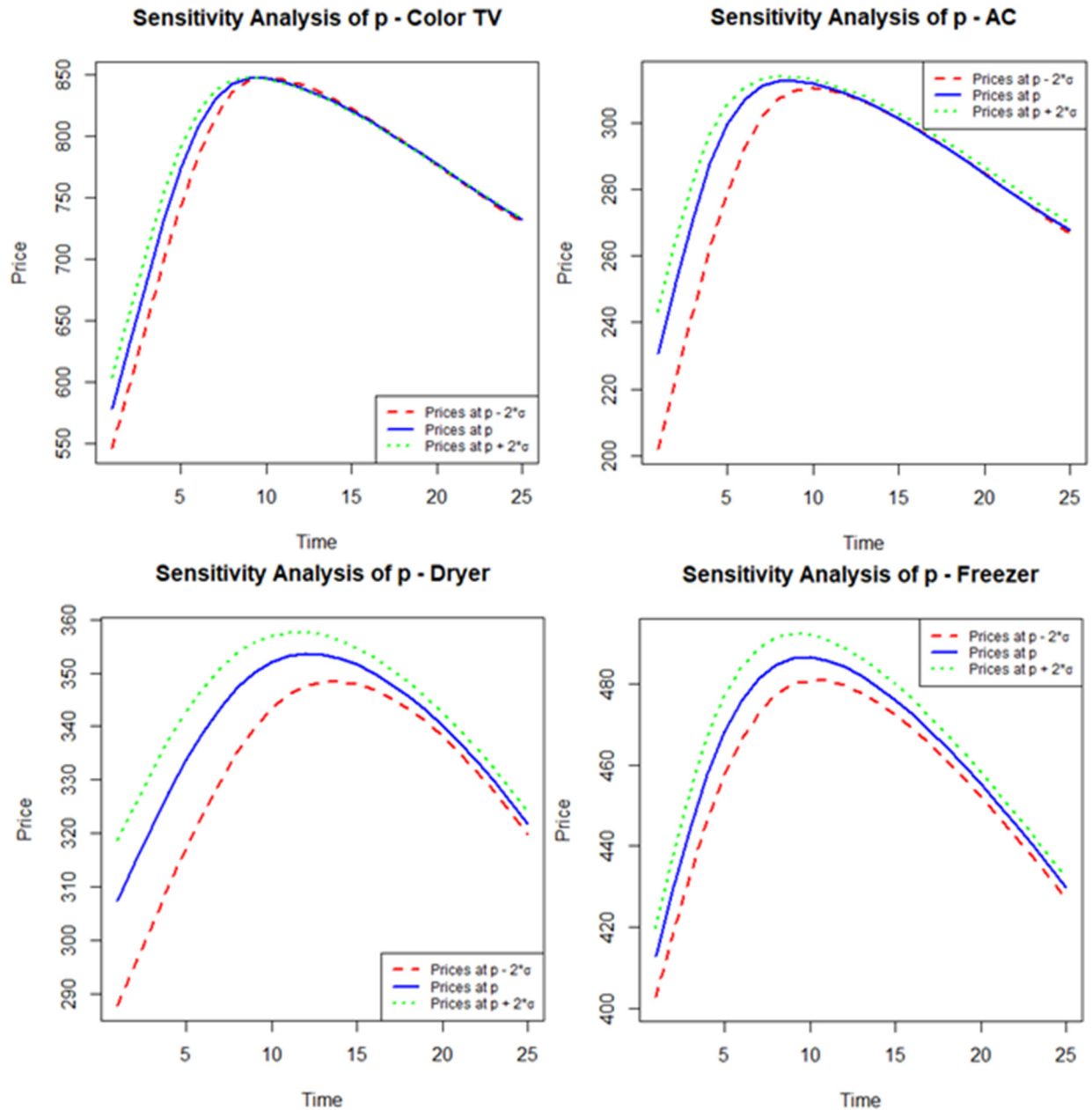
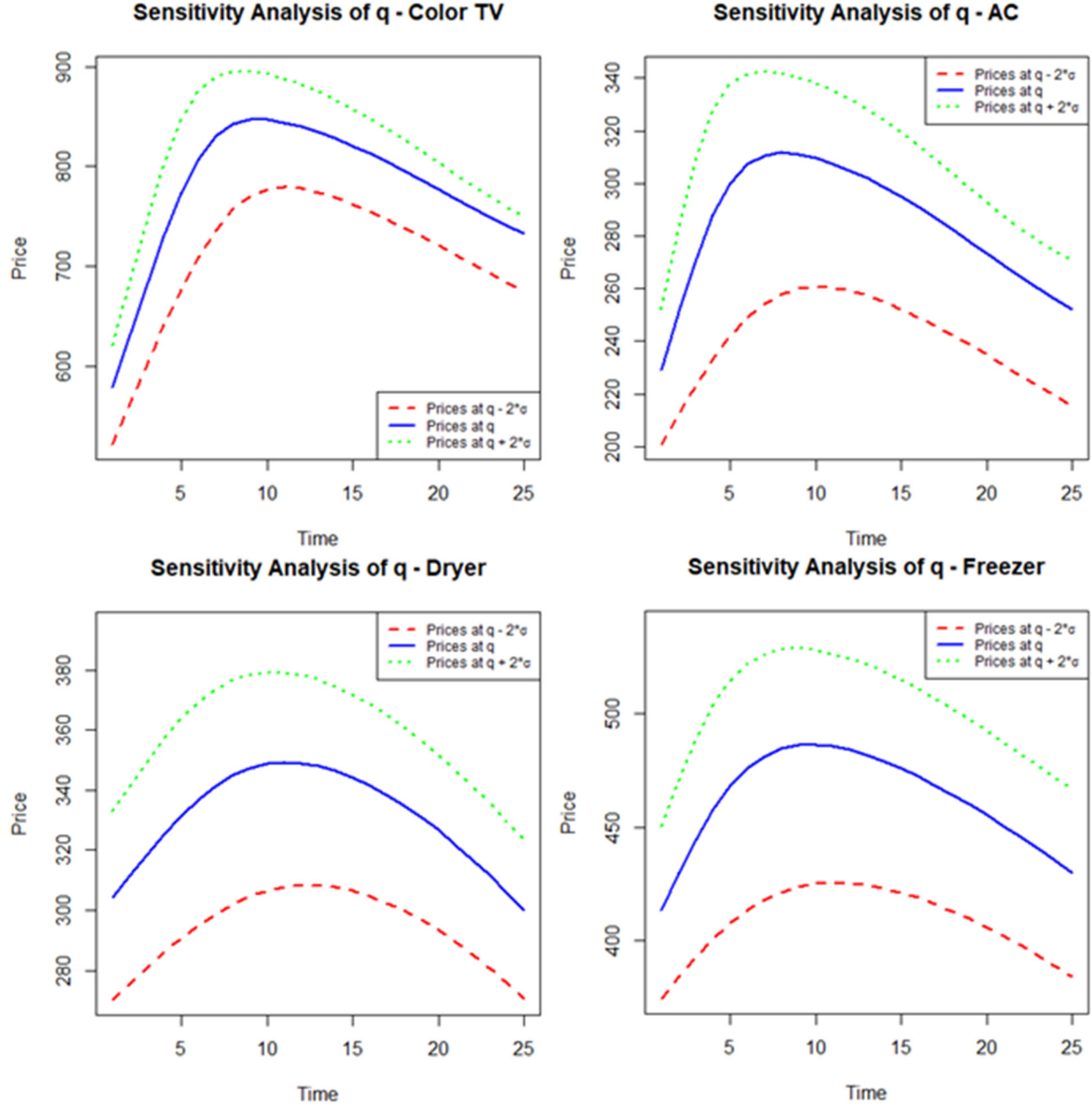


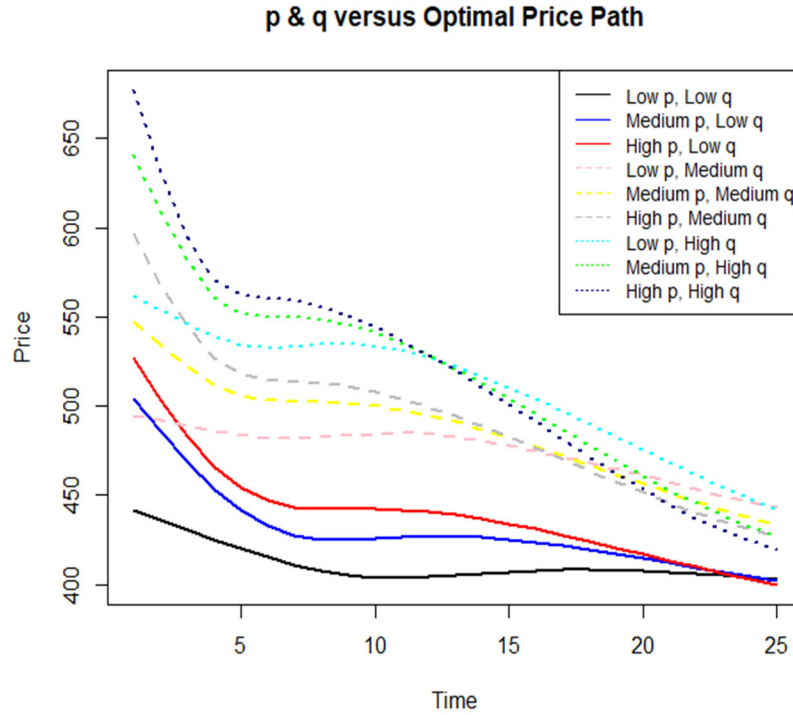
FIGURE E2: SENSITIVITY ANALYSES OF OPTIMAL PRICE PATHS WRT q



Web Appendix F: The Effect of Declining Costs on Optimal Price Paths

To calculate the dynamic price paths, rather than making any assumptions about the unobserved production costs, we allow the firm to maximize its product's lifetime revenue at the given values of the category's diffusion parameters (i.e., p and q). Our simulation studies show that optimal prices follow a non-monotonic path over time (i.e., initially increasing but then decreasing) across all combinations of p and q . On the other hand, if profit maximization is the firm's objective and the firm has a steep experiential learning curve for

the marginal cost of production for its new product, it might be possible to obtain optimal prices that fall over time. To investigate this possibility, we conduct additional simulation studies, where we assume declining costs over time or profit margin (m) increases over time as follows: $m = 1 - [0.9 / (t^{0.1})]$. Under this assumption, we obtain the following dynamic price paths for the new product. As conjectured, the optimal prices decrease overtime for all possible values of p and q .



Web Appendix G: Numerical Optimization Algorithm for the Dynamic Pricing Game

The pricing objective of the innovator firm $j=1,2$ selling the new product $j=1,2$ is shown below.

$$\{Price_{j1}, Price_{j2}, \dots, Price_{jT}\} = \operatorname{argmax} \sum_{t=1}^T \rho^{t-1} * Price_{jt} * N_{jt}$$

where ρ is the time discount factor, $Price_{jt}$ is the price of product j at time t , and N_{jt} is the expected sales of product j at time t . In this setting, each firm's pricing decision at a period affects not only its and its competitor's revenue in that period but also the revenue stream of both firms in the future (through changing the remaining market potential for new products and the rate at which consumers adopt the products in the future). The relevant state variables (Z) for the innovator firms are, therefore, time and the remaining market potential, i.e., $Z_t = \{t, R_t\}$, where $R_t = M - \sum_{j=1}^2 \sum_{k=1}^{t-1} N_{jk}$. To find the optimal pricing policy functions of each firm, we adopt a modified Pakes and McGuire (1994) algorithm that utilizes the dynamic

optimization procedure developed in Cosguner, Chan and Seetharaman (2018). First, we parameterize firm j 's temporal pricing policy as a flexible function of the state variables, Z , as shown below.

$$\text{Price}_j = \widehat{\text{Price}}(Z|\theta_j),$$

where $\widehat{\text{Price}}_1(\cdot)$ and $\widehat{\text{Price}}_2(\cdot)$ denote a flexible parametric approximation of the pricing policy of firm 1 and 2, respectively, and θ_1 and θ_2 are vectors of parameters to be estimated. To find the true θ_1 (θ_2) for a given set of diffusion parameters and the discount factor, i.e., $p, q, M, \beta, \lambda_1, \lambda_2$, and ρ , we numerically search for θ_1 (θ_2) that i) maximizes firm 1's (2's) lifetime product revenue, and ii) is the best response to firm 2's (1's) dynamic pricing policy. The algorithm is as follows.

1. Start with initial pricing vectors, $\text{Price}_1^0 = \{\text{Price}_{11}^0, \text{Price}_{12}^0, \dots, \text{Price}_{1T}^0\}$ and $\text{Price}_2^0 = \{\text{Price}_{21}^0, \text{Price}_{22}^0, \dots, \text{Price}_{2T}^0\}$.
 - 1.1. Given Price_2^0 , get Price_1^1 by using the subroutine 1.
 - 1.2. Given Price_1^1 , get Price_2^1 by using the subroutine 2.
 - 1.3. Repeat 1.1-1.2 until $\|\text{Price}^n - \text{Price}^{n-1}\| \approx 0$, where $\text{Price}^n = \begin{bmatrix} \text{Price}_1^n \\ \text{Price}_2^n \end{bmatrix}$.
 - 1.4. Set $\text{Price}^* = \text{Price}^n$

Subroutine 1: Finding Firm 1's Dynamic Best Response

The objective of this subroutine is to find the dynamic best response of firm 1 (i.e., θ_1) given a policy function of firm 2 (i.e., θ_2). Accordingly, θ_1 can be estimated via using a standard optimization routine by maximizing the lifetime revenue of firm 1 that can be calculated by using the forward simulation algorithm below.

To forward simulate the revenue of firm 1 for each candidate value of θ_1 :

- i. Start with the first period, i.e., $Z_1 = \{t = 1, R_1 = M\}$. Given Z_1 , calculate $\text{Price}_{11} = \widehat{\text{Price}}_1(Z_1|\theta_1)$ and $\text{Price}_{21} = \widehat{\text{Price}}_2(Z_1|\theta_2)$. Given $\text{Price}_{11}, \text{Price}_{21}$ and Z_1 , calculate $N_{11}(\text{Price}_{11}, \text{Price}_{21}, Z_1)$ and $N_{21}(\text{Price}_{11}, \text{Price}_{21}, Z_1)$ using equation (26).
- ii. Given $Z_2 = \{t = 2, R_2 = M - N_{11} - N_{21}\}$, calculate $\text{Price}_{12} = \widehat{\text{Price}}_1(Z_2|\theta_1)$ and $\text{Price}_{22} = \widehat{\text{Price}}_2(Z_2|\theta_2)$. Given $\text{Price}_{12}, \text{Price}_{22}$ and Z_2 , calculate $N_{12}(\text{Price}_{12}, \text{Price}_{22}, Z_2)$ and $N_{22}(\text{Price}_{12}, \text{Price}_{22}, Z_2)$ using equation (26).
- iii. Repeat (ii) until an iteration T such that $R_T = M - \sum_{j=1}^2 \sum_{k=1}^{T-1} N_{jk} \approx 0$.
- iv. Calculate the discounted sum of revenues for firm 1, i.e., $\sum_{t=1}^T \rho^{t-1} * \text{Price}_{1t} * N_{1t}$. That will be the lifetime revenue for the candidate value of θ_1 .

Subroutine 2: Finding Firm 2's Dynamic Best Response

The objective of this subroutine is to find the dynamic best response of firm 2 (i.e., θ_2) given a policy function of firm 1 (i.e., θ_1). Accordingly, θ_2 can be estimated via using a standard optimization routine by maximizing the lifetime revenue of firm 2 that can be calculated by using the forward simulation algorithm below.

To forward simulate the revenue of firm 2 for each candidate value of θ_2 :

- i. Start with the first period, i.e., $Z_1 = \{t = 1, R_1 = M\}$. Given Z_1 , calculate $Price_{11} = \widehat{Price}_1(Z_1|\theta_1)$ and $Price_{21} = \widehat{Price}_2(Z_1|\theta_2)$. Given $Price_{11}, Price_{21}$ and Z_1 , calculate $N_{11}(Price_{11}, Price_{21}, Z_1)$ and $N_{21}(Price_{11}, Price_{21}, Z_1)$ using equation (26).
- ii. Given $Z_2 = \{t = 2, R_2 = M - N_{11} - N_{21}\}$, calculate $Price_{12} = \widehat{Price}_1(Z_2|\theta_1)$ and $Price_{22} = \widehat{Price}_2(Z_2|\theta_2)$. Given $Price_{12}, Price_{22}$ and Z_2 , calculate $N_{12}(Price_{12}, Price_{22}, Z_2)$ and $N_{22}(Price_{12}, Price_{22}, Z_2)$ using equation (26).
- iii. Repeat (ii) until an iteration T such that $R_T = M - \sum_{j=1}^2 \sum_{k=1}^{T-1} N_{jk} \approx 0$.
- iv. Calculate the discounted sum of revenues for firm 2, i.e., $\sum_{t=1}^T \rho^{t-1} * Price_{2t} * N_{2t}$. That will be the lifetime revenue for the candidate value of θ_2 .